

Phase Manifold Model — First Principles and Mathematics

Time is angular, space is emergent, and depth is different

Scott Gehring

August 18, 2026

34 min read

.

(Version 2.7.0)

The deepest ingredient of physics is neither mass-energy, spacetime, nor quantum fields.

It is the continual conversion of unrealized possibilities into persistent records through contact, traversal, and closure.

Reality evolves through a three-phase structure — Before (B), During (D), and After (A).

Time is angular, space is emergent, and gravity begins inside each FTU as the time gradient $F_{\text{tu}}^G = \nabla(F_{\text{tu}}^T)$.

G is the universal gravitational crease governing the boundary relationship between interacting FTUs; θ^G is its angular measure. Contact couples their c^1 -order During contributions, establishing the per-vantage c^1/c^2 geometry through relational realization. Its angular projections enter both rail-normal depth generation and relational gravity, while the Planck-mass scale $\kappa_G = 4\pi G^N$ supplies its measurable strength.

Multi-FTU contact expresses that gradient through bond density and relational direction, with completed gravitational history recorded into Φ and geometrically modulated by δ .

1 PART I — SINGLE-FTU VANTAGE

This part develops the FTU from a single vantage. Every quantity here is evaluable from one FTU's own readouts. Contact-generated fields (Φ , δ) and the relational realization of c^2 appear here as given; their origin is defined in Part II.

1.1 Local-Vantage Convention

Within Part I, unindexed quantities belong to the active FTU vantage: $m \equiv m_i$, $d \equiv d_i$, $L_0 \equiv L_{0i}$, $S^D \equiv S_i^D$, and $z_{\text{mag}} \equiv z_{\text{mag}}^{(i)}$. Indices are used when multiple FTU vantages appear together.

1.2 1. First Principles

This section defines the B/D/A grammar, the c-scale ladder, and the base FTU coordinate map.

1.2.1 The Field Transition Unit (FTU)

The Field Transition Unit (FTU) is the fundamental Before / During / After construct through which physical reality is realized and recorded.

The Field Transition Unit (FTU) Basic Construct

$$F_{\text{tu}} = B \circ D \circ A$$

Time as BDA heuristic (ordered traversal)

$$T \approx B \circ D \circ A$$

Constants as Geometric Creases and Boundary Relations

A fundamental constant is a geometric crease expressing an invariant boundary relationship.

$$\hbar = B - D \text{ crease}$$

$$k_{\text{B}} = D - A \text{ crease}$$

$$G = \text{FTU} - \text{to} - \text{FTU gravitational crease}$$

$$c = \text{speed of light } c \approx 3.00 \times 10^8 \frac{m}{s}$$

$$m_p = \text{Planck mass } m_p \approx 2.18 \times 10^{-8} \text{ kg}$$

$$k_{\text{B}} = \text{Boltzmann constant } k_{\text{B}} \approx 1.38 \times 10^{-23} \frac{J}{K}$$

c-Powers

$$c = 299,792,458$$

$$c^2 = 89,875,517,873,681,764$$

$$c^3 = 26,944,002,417,373,989,539,335,912$$

c-Powers as Geometric and Physical Ladder

$$c^0 \rightarrow c^1 \rightarrow c^2 \rightarrow c^3$$

$$c^0 = \text{possibility point / scalar seed}$$

$$c^1 = \text{traversal edge / individual During contribution}$$

$$c^2 = \text{realized surface / relational realization (relational origin defined in Part II)}$$

$$c^3 = \text{closure volume / recorded persistence}$$

The Before Axis (Wave)

$$B = (c^0)_{\text{mp}}$$

The During Axis (Particle)

$$D = (c^1, c^2)_{\text{C}}$$

The After Axis (Cube)

$$A = (c^3)_{k_{\text{B}}}$$

FTU is a wave, particle, and cube

$$N = \text{Number of FTU Closures}$$

$$N = 1 : \text{wave}(B) \rightarrow \text{particle}(D) \rightarrow \text{cube}(A)$$

$N > 1$: interference is across A-records, rather than inside a single FTU

FTU as Coordinate System

$$F_{\text{tu}}^{\text{c}} = (B, D, A) = ((c^0)_{\text{mp}}, (c^1, c^2)_{\text{C}}, (c^3)_{k_{\text{B}}})$$

FTU as Cube

$$S = E \cdot T = \text{action}$$

$$F_{\text{tu}}^{\text{s}} = S^{\text{B}} \times S^{\text{D}} \times S^{\text{A}}$$

$$S^B = 0 \vee S^D = 0 \vee S^A = 0 \Rightarrow F_{tu}^s = 0 \Rightarrow N = 0$$

Multiplication is licensed only after the action conversion (all axes now in $\hbar$)

Cube Axes (see action readouts section)

$$S^B = (m_p c^2)(\hbar/m_p c^2) = \hbar$$

$$S^D = (mc^2)(\frac{d}{c}) = mcd$$

$$S^A = (k_B T)(\hbar/k_B T) = \hbar$$

Closure value

$$\text{On-shell } (d = \frac{\hbar}{mc}) : F_{tu}^s = \hbar^3 \text{ unit cube}$$

$$\text{Off-shell } (d \neq \frac{\hbar}{mc}) : F_{tu}^s = \hbar^2 \cdot mcd \text{ rectangular box}$$

Single FTU

$$N = F_{tu}^s/\hbar^3 = 1 \Rightarrow \text{one } \hbar^3 \text{ cube} \Leftrightarrow \text{one closure}$$

FTU Photon burst A burst of k photons is k independent closures

$$N = k \text{ The aggregate is never } (\Sigma S^B)(\Sigma S^D) \frac{(\Sigma S^A)}{\hbar^3}$$

1.3 2. Core Operators

This section defines the local phase gates, memory field, depth field, and conditioning terms used later.

1.3.1 Phase Gates

The gates organize entry, active traversal, and closure across the FTU.

Gates

$$\Theta_{in}[\Phi] = 1/(1 + e^{(-\frac{(u - u_{in})}{\sigma_{in}})})$$

$$\Theta_{\text{out}}[\diamond] = 1/(1 + e^{-(u - u_{\text{out}} - \alpha\delta)/\sigma_{\text{out}}}))$$

$\alpha > 0$ (forced by $u_{\text{in}} < u_{\text{out}}$ holding across the full δ range; large δ raises the closure threshold rather than lowering it)

$$\Theta_{\text{win}} = \Theta_{\text{in}}[\Phi](1 - \Theta_{\text{out}}[\diamond])$$

Phase progress

$$u = \text{local monotonic phase parameter } (B \rightarrow D \rightarrow A)$$

$$u_{\text{in}} < u_{\text{out}}$$

Φ Field Definition

$$\Phi = \text{single retained closure} - \text{memory field (see §7 transport law)}$$

D-phase conditioning input

$$\diamond = (F_{\text{tu}}^y, F_{\text{tu}}^x | \delta, \Theta_{\text{win}}, \Theta_{\text{con}})$$

Φ Memory Update

$$\Phi_{n+1} = \lambda \Phi_n + (1 - \lambda) S_n - \Delta\tau \nabla \cdot (\Phi_n u_n), \lambda \leftrightarrow e^{-\Delta\tau/T_p}$$

Rules

$$0 \leq \lambda \leq 1$$

$$\lambda = 0 \Rightarrow \Phi \text{ follows current FTU closure activity}$$

$$\lambda \rightarrow 1 \Rightarrow \Phi \text{ preserves prior closure geometry}$$

$$A \rightarrow \Phi$$

$$\Phi \rightarrow \delta[\Phi, \rho]$$

$$\Phi, \delta \rightarrow g_{\mu\nu}^{\text{D}}[\Phi, \delta]$$

1.4 3. FTU Spatial Behavior

This section maps D into classic x, y, and z spatial behavior, with depth active during traversal.

General idea:

During has two realized rails (x,y); depth z is generated.

$$D = (c^1, c^2)_C$$

$$c^1 \rightarrow x$$

$$c^2 \rightarrow y$$

$$C \rightarrow \text{rail gauge}$$

$$c^1 \times_r c^2 \rightarrow z_{\text{dir}}, z_{\text{mag}}$$

$$D \rightarrow (x, y, z_{\text{dir}}, z_{\text{mag}})$$

$$x_D^\mu = (ct, x, y)$$

1.4.1 XY Readouts

Xy creates the configuration surface area of the During.

Starting with the FTU as a coordinate

$$F_{\text{tu}}^c = (B, D, A)$$

Space X-axis projection

$$F_{\text{tu}}^x = (\frac{\hbar}{(m_p c)}, d, \hbar c / (k_B T)) = c \cdot F_{\text{tu}}^T$$

Space Y-axis projection

$$F_{\text{tu}}^y = (\frac{\hbar}{(m_p c)}, \frac{\hbar}{(mc)}, \hbar c / (k_B T))$$

During rail construct

$$D = (F_{\text{tu}}^x, F_{\text{tu}}^y)_C$$

Gauge

$$C = D - \text{phase rail gauge } L_0 = \frac{\hbar}{(mc)}$$

1.4.2 Depth

Depth is the D-phase rail-normal behavior of the local c^1/c^2 pair. c^1 is the active FTU's During traversal contribution; c^2 is its relational realization role. Part II defines how contact establishes that pair.

The universal gravitational crease θ_G sets the angular projection of that rail coupling. During carries the resulting temporary z-like extension, while A records the closed x-y transcription into Φ .

Depth Stack

$$z_{\text{plane}} = D \text{ rail} - \text{plane}$$

$$\times_r : (c^1, c^2) \rightarrow F_{\text{tu}}^{\text{zd}}$$

$$F_{\text{tu}}^{\text{zd}} = \text{depth readout}$$

$$z_{\text{dir}} = \text{depth direction}$$

$$z_{\text{mag}} = \text{depth magnitude} / D \text{ rail} - \text{area}$$

$$\delta = \text{depth gauge modulation}$$

$$D^\delta = \text{depth modified During}$$

Depth Axiom

$$z_{\text{plane}} = \text{span}\{c^1, c^2\}$$

$$F_{\text{tu}}^{\text{zd}} = c^1 \times_r c^2$$

$$z_{\text{dir}} = F_{\text{tu}}^{\text{zd}} / |F_{\text{tu}}^{\text{zd}}|$$

$$z_{\text{mag}} = |F_{\text{tu}}^{\text{zd}}|$$

$$F_{\text{tu}}^{\text{zd}} \rightarrow z_{\text{dir}}$$

$$F_{\text{tu}}^{\text{zd}} \neq C \neq \delta$$

z_{mag} and z_{dir} *are* independently gated :

$z_{\text{mag}} \text{---} B \rightarrow D \rightarrow A$ progress (0 in B , grows in D , sealed at A); always ≥ 0

$z_{\text{dir}} \text{---} \Theta_{\text{out}}/\Theta_{\text{in}}$ dominance :

Θ_{out} dominant $\rightarrow +z_{\text{dir}}$ (outward)

Θ_{in} dominant $\rightarrow -z_{\text{dir}}$ (inward)

Rail Spans

$$c^1 \rightarrow x \text{--- rail}, F_{\text{tu}}^{\text{xd}} = d c^2 \rightarrow y \text{--- rail}, F_{\text{tu}}^{\text{yd}} = L_0 L_0 = \frac{\hbar}{(mc)}$$

Gravitational-Crease Depth Projection

$$\theta_G = 45^\circ$$

$$z_{\text{mag}} = F_{\text{tu}}^{\text{xd}} F_{\text{tu}}^{\text{yd}} \sin(\theta_G)$$

$$z_{\text{mag}} = d L_0 \sin(\theta_G)$$

$$z_{\text{mag}} = d \cdot \frac{\hbar}{(mc)} \cdot \sin(\theta_G)$$

On Shell:

$$d = L_0 = \frac{\hbar}{(mc)} z_{\text{mag}} = L_0^2 \sin(\theta_G) z_{\text{mag}} = L_0^2 / \sqrt{2}$$

Readout Rule

$$F_{\text{tu}}^{\text{y}} = Y[x^\mu]$$

$$F_{\mathrm{tu}}^{\mathrm{x}} = X[x^{\mu}]$$

$$F_{\mathrm{tu}}^{\mathrm{zd}} = c^1 \times_r c^2$$

$$D = (F_{\mathrm{tu}}^{\mathrm{xd}}, F_{\mathrm{tu}}^{\mathrm{yd}}, F_{\mathrm{tu}}^{\mathrm{zd}})$$

Phase Behavior

$$B : \text{rails unopened} \Rightarrow z_{\text{mag}} = 0$$

$$D : \text{rails differentiate} \Rightarrow z_{\text{mag}} > 0$$

$$A : \Theta_{\text{out}} \text{ closes} \Rightarrow \mathcal{T}_A[z_{\text{mag}}/A_{\text{ref}}] \rightarrow A = (c^3)_{k_{\text{B}}}$$

$$A_{\text{ref}} = 4\ell_P^2$$

Depth Rule

$$\delta \notin B$$

$$\delta \in D$$

$$\delta \notin A$$

Depth field

$$\delta = \delta[\Phi,\rho]$$

$$\delta[\Phi,\rho] = \ell_\delta \cdot |\nabla^2\Phi|/(|\nabla\Phi| + \epsilon_\Phi) \cdot \frac{\rho}{(\rho + \rho_c)}$$

Depth-modified FTU

$$F_{\mathrm{tu}}^\delta = B \circ D^\delta \circ A$$

$$D^\delta = (c^1,c^2)_{\mathcal{C}^\delta}$$

GR limit

$\delta \rightarrow 0$ leaves z , z_{mag} , and $\mathbf{C}$ intact

$$g_{\mu\nu}^{\text{D}}(\Phi, 0) = g_{\mu\nu}^{\text{GR}}(\Phi)$$

Notes on depth stack

In this model, z , z_{mag} , and δ are separate roles: z is the D-phase depth readout, z_{mag} is the rail-area magnitude, and δ is the field modulation that changes how depth affects the D-metric.

1.4.3 Spatial Coordinate Readout Summary

This section unifies the XY and depth readouts into what is perceived as space in the During.

Definition

$$z = \text{depth perception}$$

Starting with During Frame

$$D = (c^1, c^2)_{\mathbf{C}}$$

$$c^1 \rightarrow F_{\text{tu}}^{\text{xd}}$$

$$c^2 \rightarrow F_{\text{tu}}^{\text{yd}}$$

$$\mathbf{C} \rightarrow D - \text{phase gauge}$$

$$F_{\text{tu}}^{\text{xd}} = d F_{\text{tu}}^{\text{yd}} = L_0 L_0 = \frac{\hbar}{(mc)}$$

$$F_{\text{tu}}^{\text{zd}} = c^1 \times_r c^2$$

$$F_{\text{tu}}^{\text{zd}} \rightarrow x \times y \rightarrow z$$

$$z = F_{\text{tu}}^{\text{zd}} D = (d, L_0; z_{\text{dir}}, z_{\text{mag}})$$

Perceived space

$$D \rightarrow (F_{\text{tu}}^{\text{xd}}, F_{\text{tu}}^{\text{yd}}, F_{\text{tu}}^{\text{zd}})$$

$$D \rightarrow (x, y; z_{\text{dir}}, z_{\text{mag}})$$

1.5 4. FTU Readouts

This section collects the physical readouts produced by the FTU across B, D, and A.

1.5.1 Primary Readouts

FTU Readouts are the measurable manifestations of an FTU expressed in a specific physical representation.

Starting with the FTU as a coordinate

$$F_{\text{tu}}^{\text{c}} = (B, D, A)$$

Mass readout

$$\text{Mass} = F_{\text{tu}}^{\text{M}} = (m_p, m, \frac{(k_{\text{B}}T)}{c^2})$$

Energy readout

$$\text{Energy} = F_{\text{tu}}^{\text{E}} = (m_p c^2, m c^2, k_{\text{B}}T)$$

Time readout

$$\text{Time} = F_{\text{tu}}^{\text{T}} = (\frac{\hbar}{(m_p c^2)}, \frac{d}{c}, \frac{\hbar}{(k_{\text{B}}T)})$$

Gravity readout

$$\text{Gravity} = F_{\text{tu}}^{\text{G}} = (\nabla(\frac{\hbar}{(m_p c^2)}), \nabla(\frac{d}{c}), \nabla(\frac{\hbar}{(k_{\text{B}}T)}))$$

1.5.2 Motion Readouts

Motion Readouts collect the D-phase quantities that describe traversal through the realized x-y surface.

Velocity readout

$$F_{\text{tu}}^{\text{V}} = (c, v, c)$$

Force readout

$$F_{\text{tu}}^{\text{f}} = (\Delta p^{\text{B}}/\Delta t^{\text{B}}, \Delta p^{\text{D}}/\Delta t^{\text{D}}, \Delta p^{\text{A}}/\Delta t^{\text{A}})$$

D-Phase Force Recovery

$$F^{\text{D}} = dp^{\text{D}}/dt^{\text{D}}$$

Momentum readout

$$F_{\text{tu}}^{\text{P}} = (m_p c, p, k_{\text{B}} \frac{T}{c})$$

1.5.3 Before Only Readouts

Before Only Readouts are observables extracted solely from the potential phase B.

Coordinate Projections (isotropic)

$$F_{\text{tu}}^{\text{yb}} = \frac{\hbar}{(m_p c)}$$

$$F_{\text{tu}}^{\text{xb}} = \frac{\hbar}{(m_p c)}$$

Readouts

$$F_{\text{tu}}^{\text{Gb}} = \nabla(\frac{\hbar}{(m_p c^2)})$$

$$F_{\text{tu}}^{\text{Tb}} = \frac{\hbar}{(m_p c^2)}$$

$$F_{\text{tu}}^{\text{Eb}} = m_p c^2$$

$$F_{\text{tu}}^{\text{Mb}} = m_p$$

$$F_{\text{tu}}^{\text{Vb}} = c$$

$$F_{\text{tu}}^{\text{fb}} = \Delta p^{\text{B}}/\Delta t^{\text{B}}$$

$$F_{\text{tu}}^{\text{Pb}} = m_p c$$

1.5.4 During Only Readouts

During Only Readouts are observables extracted solely from the realized geometric phase D.

Traversal Coordinate Projection

$$F_{\text{tu}}^{\text{xd}} = d$$

Realization Coordinate Projection

$$F_{\text{tu}}^{\text{yd}} = \frac{\hbar}{(mc)} = L_0$$

Readouts

$$F_{\text{tu}}^{\text{Gd}} = \nabla\left(\frac{d}{c}\right) F_{\text{tu}}^{\text{Td}} = \frac{d}{c} F_{\text{tu}}^{\text{Ed}} = mc^2 F_{\text{tu}}^{\text{Md}} = m F_{\text{tu}}^{\text{Vd}} = v F_{\text{tu}}^{\text{fd}} = \Delta p^{\text{D}} / \Delta t^{\text{D}} F_{\text{tu}}^{\text{Pd}} = p$$

1.5.5 After Only Readouts

After Only Readouts are observables extracted solely from the persistent informational phase A.

Coordinate Projections (isotropic)

$$F_{\text{tu}}^{\text{ya}} = \hbar c / (k_{\text{B}} T)$$

$$F_{\text{tu}}^{\text{xa}} = \hbar c / (k_{\text{B}} T)$$

Readouts

$$F_{\text{tu}}^{\text{Ma}} = \frac{(k_{\text{B}} T)}{c^2}$$

$$F_{\text{tu}}^{\text{Ea}} = k_{\text{B}} T$$

$$F_{\text{tu}}^{\text{Ta}} = \frac{\hbar}{(k_{\text{B}} T)}$$

$$F_{\text{tu}}^{\text{Ga}} = \nabla\left(\frac{\hbar}{(k_{\text{B}} T)}\right)$$

$$F_{\text{tu}}^{\text{Va}} = c$$

$$F_{\text{tu}}^{\text{Pa}} = k_{\text{B}} \frac{T}{c}$$

$$F_{\text{tu}}^{\text{fa}} = \Delta p^{\text{A}} / \Delta t^{\text{A}}$$

1.5.6 Action Readouts

FTU Action Readouts are derived from action = energy $\times$ time ($S = E \cdot T$) and allow a standard unit of measure across the BDA domains, allowing the FTU to function as a cube.

Action

$$S = E \cdot T$$

B-axis action readout

$$S^{\text{B}} = F_{\text{tu}}^{\text{Eb}} \cdot F_{\text{tu}}^{\text{Tb}}$$

$$S^{\text{B}} = (m_p c^2) \left(\frac{\hbar}{(m_p c^2)} \right)$$

$$S^{\text{B}} = \hbar$$

D-axis action readout

$$S^{\text{D}} = F_{\text{tu}}^{\text{Ed}} \cdot F_{\text{tu}}^{\text{Td}} S^{\text{D}} = (mc^2) \left(\frac{d}{c} \right) S^{\text{D}} = mcd$$

A-axis action readout

$$S^{\text{A}} = F_{\text{tu}}^{\text{Ea}} \cdot F_{\text{tu}}^{\text{Ta}}$$

$$S^{\text{A}} = (k_{\text{B}} T) \left(\frac{\hbar}{(k_{\text{B}} T)} \right)$$

$$S^{\text{A}} = \hbar$$

FTU Action Readout

$$S^{\text{B}} = F_{\text{tu}}^{\text{Eb}} \cdot F_{\text{tu}}^{\text{Tb}} = \hbar S^{\text{D}} = F_{\text{tu}}^{\text{Ed}} \cdot F_{\text{tu}}^{\text{Td}} = mcd S^{\text{A}} = F_{\text{tu}}^{\text{Ea}} \cdot F_{\text{tu}}^{\text{Ta}} = \hbar$$

FTU Cube based on action

$$F_{\text{tu}}^{\text{s}} = S^{\text{B}} \times S^{\text{D}} \times S^{\text{A}}$$

$$S^{\text{B}} = 0 \vee$$

$$S^D = 0 \vee$$

$$S^A = 0 \Rightarrow$$

$$F_{\text{tu}}^s = 0 \Rightarrow N = 0$$

1.5.7 Entropy Readout

The entropy readout tracks how the FTU possibility becomes a conditioned probability and then a recorded A-phase information value.

FTU Entropy Readout

$$F_{\text{tu}}^H = (H^B, H^D, H^A)$$

Before entropy

$$H^B = -\sum_i p_i^0 \log_2 p_i^0$$

During conditioned entropy

$$H^D = -\sum_i P(\phi_i \mid \diamond) \log_2 P(\phi_i \mid \diamond)$$

After record information

$$H^A = -\log_2 p_K$$

Meaning

$$H^B = \text{prior possibility entropy}$$

$$H^D = \text{conditioned closure entropy}$$

$$H^A = \text{recorded } A - \text{entropy of the selected closure}$$

A-phase record

$$A \rightarrow \phi_K$$

$$\phi_K \rightarrow H^A$$

$$\phi_K \rightarrow S_n$$

$$S_n \rightarrow \Phi_{n+1}$$

1.6 5. During Geometry

This section turns the D-phase rail system into metric, geodesic, and motion language.

1.6.1 Metric

The FTU metric is the effective spacetime geometry induced by D and shaped by Φ and δ .

Metric Definitions

$$g^D = D - \text{phase metric geometry}$$

$$g^{\text{GR}} = \text{General Relativity metric geometry}$$

$$g_{\mu\nu}^D[\Phi, \delta] = D - \text{phase metric generated by } \Phi \text{ and modulated by } \delta$$

$$g_{\mu\nu}^{\text{GR}}[\Phi] = \text{GR} - \text{limit metric generated by } \Phi$$

$$h_{\mu\nu}^\delta[\delta] = \text{pure depth correction}$$

$$h_{\mu\nu}^{\Phi\delta}[\Phi, \delta] = \text{memory} - \text{depth coupling correction}$$

D opens the coupled During geometry

$$D = (c^1, c^2)_C$$

Relational origin of c^2

Defined in Part II. Part I uses the established local pair $D = (c^1, c^2)_C$.

Functional rail readout

$$c^1 \rightarrow \text{traversal role}$$

$$c^2 \rightarrow \text{relational realization role}$$

xy construct

$$D = (F_{\text{tu}}^x, F_{\text{tu}}^y)_{\mathcal{C}}$$

c^1 rail carries time/gradient behavior

$$c^1 \rightarrow (F_{\text{tu}}^{\text{Td}}, F_{\text{tu}}^{\text{Gd}})$$

c^2 rail carries energy/mass behavior

$$c^2 \rightarrow (F_{\text{tu}}^{\text{Ed}}, F_{\text{tu}}^{\text{Md}})$$

$\mathcal{C}$ sets the light-speed causal gauge

$$\mathcal{C} \rightarrow c$$

Depth-modified During

$$D^\delta = (c^1, c^2)_{\mathcal{C}^\delta}$$

$$c^1 \rightarrow x$$

$$c^2 \rightarrow y$$

$$z = c^1 \times_r c^2$$

$$D^\delta \rightarrow (x, y; z_{\text{dir}}, z_{\text{mag}})$$

Depth-modified During with During readouts

$$D^\delta = (c^1_{(F_{\text{tu}}^{\text{Td}}, F_{\text{tu}}^{\text{Gd}})}, c^2_{(F_{\text{tu}}^{\text{Ed}}, F_{\text{tu}}^{\text{Md}})})_{\mathcal{C}^\delta}$$

During-to-Metric Projection

$$D^\delta = (c^1, c^2)_{\mathcal{C}^\delta} \rightarrow (x, y; z_{\text{dir}}, z_{\text{mag}})$$

$$\rightarrow x_D^\mu = (ct, x, y)$$

$$z = c^1 \times_r c^2 \text{ generated (rail} - \text{normal),}$$

not a metric coordinate; D-metric is 2+1 over (ct,x,y)

$$\rightarrow ds_D^2 = g_{\mu\nu}^D[\Phi, \delta] dx_D^\mu dx_D^\nu$$

Shorthand

$$D^\delta \Rightarrow ds_D^2$$

GR Limit

$$g_{\mu\nu}^D[\Phi, 0] = g_{\mu\nu}^{\text{GR}}[\Phi]$$

Metric Decomposition

$$g_{\mu\nu}^D[\Phi, \delta] = g_{\mu\nu}^{\text{GR}}[\Phi] + h_{\mu\nu}^\delta[\delta] + h_{\mu\nu}^{\Phi\delta}[\Phi, \delta]$$

PMM Spacetime Line Element

$$ds^2 = \Theta_{\text{win}} \cdot g_{\mu\nu}^D[\Phi, \delta] dx_D^\mu dx_D^\nu$$

$$\text{GR/exterior limit : } \Theta_{\text{win}} \rightarrow 1 \Rightarrow ds^2 \rightarrow g_{\mu\nu}^D[\Phi, \delta] dx_D^\mu dx_D^\nu$$

Phase-Time Readout

$$d\tau_{\text{eff}}^2 = w_B d\tau_B^2 + w_D d\tau_D^2 + w_A d\tau_A^2$$

1.6.2 Lagrangian

The PMM Lagrangian states how recorded transition memory Φ and depth δ generate the D-phase geometry through action variation.

PMM D-Phase Action

$$\mathcal{A}_{\text{field}}^D = \int d^4x \sqrt{-g^D} \mathcal{L}^D$$

D-Phase Lagrangian Density

$$\mathcal{L}^D = (\frac{c^4}{16} \pi G_N) R[g^D] \mathcal{L}_\Phi \mathcal{L}_\delta \mathcal{L}_{\text{int}}[\Phi, \delta] \mathcal{L}_m$$

Depth Dynamics

$$\mathcal{L}_\delta = -(\frac{1}{2})\Theta_{\text{win}}(\partial_\mu \delta)(\partial^\mu \delta) - \frac{\alpha_\delta}{2}\delta^2 \cdot \frac{\rho_c}{(\rho + \rho_c)}$$

Field Variation

$$\delta \mathcal{A}_{\text{field}}^{\text{D}} / \delta g_{\mu\nu}^{\text{D}} \rightarrow D - \text{phase field equations}$$

$$\delta \mathcal{A}_{\text{field}}^{\text{D}} / \delta \delta \rightarrow \text{depth field equation}$$

$$\delta \mathcal{A}_{\text{field}}^{\text{D}} / \delta \Phi \rightarrow \text{memory} - \text{gradient field equation}$$

Φ -Sector

$\mathcal{L}_{\Phi}$ generates the Φ transport/persistence law :

$$\partial \Phi / \partial \tau_A + \nabla \cdot (\Phi u) = \Gamma_{\text{@}}^{\Phi} \Theta_{\text{out}} (\Delta T)_{\text{eff}}^2 - \Phi / T_p$$

stated as field equation; variational $\mathcal{L}_{\Phi}$ form still open

$\mathcal{L}_{\text{int}}[\Phi, \delta]$ reserved for the $\Phi \delta$ – coupling sector.

1.6.3 D-Phase Action Momentum Scale

The D-phase action momentum scale derives from During action.

D-phase action momentum scale

$$p_{\text{action}}^{\text{D}} = S^{\text{D}} / d S^{\text{D}} = m c d p_{\text{action}}^{\text{D}} = m c$$

Action relation

$$S^{\text{D}} = p_{\text{action}}^{\text{D}} d$$

Massive relativistic momentum

$$\chi(v) = \sqrt{1 - \frac{v^2}{c^2}}$$

$$\gamma(v) = \frac{1}{\chi}(v)$$

$$p_{\text{xy}}(v) = \gamma m v_{\text{xy}} \quad 0 \leq v < c$$

Massless light momentum

$$m = 0$$

$$p\gamma = E\frac{\gamma}{c}$$

1.6.4 Momentum–Kinematics–Metric Bridge

This bridge separates translational momentum, During-flow velocity, and rail-normal carrier participation energy.

Metric path

$$D^\delta \rightarrow g_{\mu\nu}^{\text{D}}[\Phi, \delta]$$

$$x_D^\mu = (ct, x, y)$$

$$z \text{ is rail - normal } (c^1 \times_r c^2)$$

$$p_{\text{xy}} \in \text{span}\{x,y\}$$

$$M_{\text{op}}^{\text{c}} \parallel z_{\text{dir}}$$

$$M_{\text{op}}[D^\delta] \in T(\Pi_{\text{xy}})$$

Geodesic tangent

$$u_D^\mu = \frac{dx_D^\mu}{d\tau^{\text{D}}}$$

Momentum along the path

$$p_D^\mu = m\,u_D^\mu$$

$$\text{spatial part} \in \text{span}\{x,y\} \text{ (see } D - \text{Velocity Rule)}$$

$$\Rightarrow p_{\text{xy}} = \gamma\,m\,v_{\text{xy}}, \text{ no } z \text{ component}$$

$$\textbf{Momentum Exchange } p_{\text{carrier}} \in F_{\text{tu}}^{\text{P}} \; p_{1i}, p_{1f} \in p_{\text{xy}}^1 \; p_{2i}, p_{2f} \in p_{\text{xy}}^2 \; q_1 = p_1 f - p_{1i} \; q_2 = p_2 f - p_{2i} \\ q_1 + q_2 + p_{\text{carrier}} = \mathbf{0}$$

If no emitted carrier $p_{\text{carrier}} = 0$ $q_1 + q_2 = 0$

Massive spatial momentum

$$p_{\text{xy}}(v) = \gamma m v_{\text{xy}}$$

During-Flow Velocity Operator

$$M_{\text{op}}[D^\delta] = v_{\text{flow}}^{\text{D}}$$

$$\text{massive carrier} : v_{\text{flow}}^{\text{D}} = v_{\text{xy}}$$

$$\text{photon} : M_{\text{op}}[D^{e\delta}] = c \hat{t}^{\text{D}}, v_{\text{xy}} \text{ undefined}$$

$$[M_{\text{op}}[D^\delta]] = L \cdot T^{-1} = m \cdot s^{-1}$$

Carrier Participation Energy Readout

$$M_{\text{op}}^{\text{c}} = \chi(v) \cdot E_{\text{realization}} \cdot z_{\text{dir}}$$

$$[M_{\text{op}}^{\text{c}}] = E = J = kg \cdot m^2 \cdot s^{-2}$$

$$^{\text{c}} = \text{carrier}$$

Interpretation

g^{D} sets the path. $\hat{u}_{\mu_D}$ sets the path direction. $\hat{p}_{\mu_D}$ gives the momentum carried along that path. M_{op} maps a During propagation mode to a velocity readout. M_{op}^{c} gives the rail-normal carrier participation energy.

1.6.5 Geodesic

The geodesic block describes free traversal through the D-phase metric.

Start from During Axis

$$D = (c^1, c^2)_{\text{C}}$$

D-Phase Metric w/ Depth (see gravity section for derivative)

$$D^\delta \rightarrow g_{\mu\nu}^{\text{D}}(\Phi, \delta)$$

Spacetime metric

$$ds^2 = \Theta_{\text{win}} \cdot g_{\mu\nu}^{\text{D}}(\Phi, \delta) dx^\mu dx^\nu$$

D-path action

$$\mathcal{A}_D = \int ds_D$$

Free D-motion

$$\delta \mathcal{A}_D = 0$$

Geodesic

$$\frac{d^2 x^\mu}{d(\tau^{\text{D}})^2} + \Gamma_{\alpha\beta}^\mu [g^{\text{D}}(\Phi, \delta)] \cdot \left(\frac{dx^\alpha}{d\tau^{\text{D}}} \right) \left(\frac{dx^\beta}{d\tau^{\text{D}}} \right) = 0$$

FTU Readouts From Single Geodesic

$$F_{\text{tu}}^{\text{y}} = Y[x^\mu(\tau^{\text{D}})]$$

$$F_{\text{tu}}^{\text{x}} = X[x^\mu(\tau^{\text{D}})]$$

1.6.6 Kinematics

This block defines During-flow velocity, carrier participation energy, lightlike D propagation, and gate-derived rail orientation.

Starting from During Operator Rails

$$D = (c^1, c^2)_{\text{C}}$$

$$\text{C} = c$$

Depth-modified During

$$\text{C}^\delta = \delta - \text{modulated C}$$

$$D^\delta = (c^1, c^2)_{\text{C}^\delta}$$

Kinematic Rail Definitions

$$c^1 = \text{During traversal role}$$

c^2 = relational realization role (relational origin defined in Part II)

Physical readouts

$$c^1 \rightarrow (F_{\text{tu}}^{\text{Td}}, F_{\text{tu}}^{\text{Gd}}) c^2 \rightarrow (F_{\text{tu}}^{\text{Ed}}, F_{\text{tu}}^{\text{Md}}) c^1 \rightarrow$$

$$F_{\text{tu}}^{\text{xd}} = d c^2 \rightarrow$$

$$F_{\text{tu}}^{\text{yd}} = L_0 E\gamma \text{ active for photons}$$

Depth-modified During with kinematic Rail Definitions

$$D^\delta = (c_{(F_{\text{tu}}^{\text{Td}}, F_{\text{tu}}^{\text{Gd}})}^1, c_{(F_{\text{tu}}^{\text{Ed}}, F_{\text{tu}}^{\text{Md}})}^2) \mathcal{C}^\delta$$

Using rail coupling ($\times_r$) to create depth readout

$$F_{\text{tu}}^{\text{zd}} = c^1 \times_r c^2$$

$$z_{\text{mag}} = |F_{\text{tu}}^{\text{zd}}|$$

$$z_{\text{dir}} = F_{\text{tu}}^{\text{zd}} / |F_{\text{tu}}^{\text{zd}}|$$

$$F_{\text{tu}}^{\text{zd}} = z_{\text{mag}} \cdot z_{\text{dir}}$$

depth readout

$$z = c^1 \times_r c^2$$

Depth readout with gauge kinematic expanded notation

$$z = c_{(F_{\text{tu}}^{\text{Td}}, F_{\text{tu}}^{\text{Gd}})}^1 \times_r c_{(F_{\text{tu}}^{\text{Ed}}, F_{\text{tu}}^{\text{Md}})}^2$$

Depth Field

$$\delta = \delta[\Phi, \rho]$$

$$\delta[\Phi, \rho] = \ell_\delta \cdot |\nabla^2 \Phi| / (|\nabla \Phi| + \epsilon_\Phi) \cdot \frac{\rho}{(\rho + \rho_c)}$$

Readouts

$$F_{\text{tu}}^{\text{Gd}} = \nabla\left(\frac{d}{c}\right) F_{\text{tu}}^{\text{Td}} = \frac{d}{c} F_{\text{tu}}^{\text{Ed}} = mc^2 F_{\text{tu}}^{\text{Md}} = m F_{\text{tu}}^{\text{xd}} = d F_{\text{tu}}^{\text{yd}} = L_0 L_0 = \frac{\hbar}{(mc)}$$

D-Velocity Rule

Realized surface

$$\Pi_{\text{xy}} = \text{span}\{x, y\}$$

Body velocity

$$v_{\text{xy}} = \frac{dx_{\text{xy}}}{dt}, \quad v_{\text{xy}} \in T(\Pi_{\text{xy}}), \quad v = |v_{\text{xy}}|, \quad 0 \leq v < c$$

light limit: $v = c$

Rail-normal depth channel

$$F_{\text{tu}}^{\text{zd}} = c^1 \times_r c^2$$

$$z_{\text{dir}} = F_{\text{tu}}^{\text{zd}} / |F_{\text{tu}}^{\text{zd}}|$$

Momentum projection

$$p_{\text{xy}}(v) = \gamma m v_{\text{xy}}$$

Carrier Participation Energy Readout

$$M_{\text{op}}^{\text{c}} = \chi(v) \cdot E_{\text{realization}} \cdot z_{\text{dir}}$$

$$[M_{\text{op}}^{\text{c}}] = E = J = kg \cdot m^2 \cdot s^{-2}$$

For light $\mathbf{v} = \mathbf{c}$ is an axiom of the light-limit, since

$$L_0 = \frac{\hbar}{(mc)} \text{ is undefined at } m = 0.$$

χ from D-gauge budget

v = carrier – to – D relative readout speed

c_C = speed – valued carrier participation readout

$$[v] = [c_C] = L \cdot T^{-1} = m \cdot s^{-1}$$

$$c^2 = v^2 + c_C^2$$

$$c_C = c\chi$$

$$\chi = c_C/c$$

$$\chi^2 = 1 - \frac{v^2}{c^2}$$

$$\chi(v) = \sqrt{1 - \frac{v^2}{c^2}}$$

$$\gamma = \frac{1}{\chi}$$

Rail Coupling

$$z_{\text{plane}} = D \text{ rail} - \text{plane}$$

$$(c^1, c^2) \in z_{\text{plane}}$$

$$\theta_G = \text{angular measure of the universal } G \text{ crease in the rail coupling } z_{\text{dir}} \perp z_{\text{plane}}$$

$$\times_r = \text{gravitational} - \text{crease rail} - \text{coupling product}$$

$$F_{\text{tu}}^{\text{zd}} = c^1 \times_r c^2$$

$$|F_{\text{tu}}^{\text{zd}}| = |c^1||c^2| \sin(\theta_G) z_{\text{mag}} = dL_0 \sin(\theta_G)$$

$$c^1 \times_r c^2 \rightarrow \text{oriented rail} - \text{normal } z_{\text{dir}}$$

$$z_{\text{dir}} = \pm \text{rail} - \text{normal}$$

Gate-Derived Orientation Readout

Θ_{out} dominant $\rightarrow +z_{\text{dir}}$ outward

Θ_{in} dominant $\rightarrow -z_{\text{dir}}$ inward

$z_{\text{dir}} = \text{gate} - \text{derived rail} - \text{normal orientation readout}$

Role in M_{op}^c

$\chi > 0 \Rightarrow z_{\text{dir}}$ orients M_{op}^c

$\chi = 0 \Rightarrow M_{\text{op}}^c = 0$

Relativistic Rail Deformation

$$D^{\gamma\delta} = (\frac{c^1}{\chi}, c^2\chi)_{\mathcal{C}^\delta} = (\gamma c^1, \frac{c^2}{\gamma})_{\mathcal{C}^\delta}$$

Gravitational-Crease Invariance

$$\theta_G(v) = \theta_G = 45^\circ$$

$$z_{\text{mag}}(v) = (\frac{d}{\chi})(L_0\chi) \sin(\theta_G) z_{\text{mag}}(v) = dL_0 \sin(\theta_G)$$

$$\Rightarrow z_{\text{mag}}(v) = z_{\text{mag}}(0)$$

The reciprocal SR rail deformation preserves the gravitational-crease depth projection.

Energy Realization Readout

$$E_{\text{realization}} \equiv F_{\text{tu}}^{\text{Ed}} \text{ massive} : E_{\text{realization}} = mc^2 \text{ photon} : E_{\text{realization}} = E\gamma[E_{\text{realization}}] = E = J = kg \cdot m^2 \cdot s^{-2}$$

Carrier Participation Energy Readout

$$M_{\text{op}}^c = \chi(v) \cdot E_{\text{realization}} \cdot z_{\text{dir}}$$

$$[M_{\text{op}}^c] = E = J = kg \cdot m^2 \cdot s^{-2}$$

Expanded Carrier Readout

$$z = c_{(F_{\text{tu}}^{\text{Td}}, F_{\text{tu}}^{\text{Gd}})}^1 \times_r c_{(F_{\text{tu}}^{\text{Ed}}, F_{\text{tu}}^{\text{Md}})}^2$$

$$|M_{\text{op}}^c| = \chi(v) \cdot E_{\text{realization}}$$

$$\text{direction} = z_{\text{dir}}$$

During Propagation Modes

$$D^\delta = \text{During propagation mode through } \mathbb{C}^\delta$$

$$D^{c\delta} = \text{lightlike limit of } D^{\gamma\delta} \text{ as } \chi \rightarrow 0$$

During-Flow Velocity Operator

$$M_{\text{op}}[D^\delta] = v_{\text{flow}}^{\text{D}}$$

$$[M_{\text{op}}[D^\delta]] = L \cdot T^{-1} = m \cdot s^{-1}$$

$$v_{\text{flow}}^{\text{D}} = \text{velocity of the } D - \text{propagation mode}$$

(During moving past the marker)

$$v_{\text{xy}} = \text{body velocity carried on } \Pi_{\text{xy}}; \text{ photon } \textit{has} \text{ none } (m = 0)$$

$$\text{massive carrier : body locks to flow} \Rightarrow v_{\text{flow}}^{\text{D}} = v_{\text{xy}}$$

$$\text{photon : pure flow, no body} \Rightarrow M_{\text{op}}[D^{c\delta}] = c \hat{t}^{\text{D}}, v_{\text{xy}} \text{ undefined}$$

$$\hat{t}^{\text{D}} = \text{unit tangent of } D - \text{flow}$$

$$[\hat{t}^{\text{D}}] = 1$$

Lightlike During Readout

$$M_{\text{op}}[D^{c\delta}] = c \hat{t}^{\text{D}}$$

Light Limit

$$v = c \Leftrightarrow \chi = 0 \Leftrightarrow c_C = 0 \Leftrightarrow M_{\text{op}}^c = 0$$

$$M_{\text{op}}[D^{c\delta}] = c \hat{t}^{\text{D}}$$

Two Zeros of Carrier Participation Energy

$$\text{rail} - \text{degenerate} : F_{\text{tu}}^{\text{zd}} = 0 \Rightarrow M_{\text{op}}^c = 0$$

$$\text{lightlike} : v = c \Rightarrow \chi = 0 \Rightarrow M_{\text{op}}^c = 0$$

Carrier Case Massive

$$0 \leq v < c \chi > 0 m > 0 E_{\text{realization}} = mc^2 |M_{\text{op}}^c| > 0 M_{\text{op}}[D^\delta] = v_{\text{flow}}^{\text{D}}$$

Carrier Case Photon

$$v = c$$

$$\chi = 0$$

$$c_C = 0$$

$$F_{\text{tu}}^{\text{Md}} = 0$$

$$F_{\text{tu}}^{\text{Ed}} = E\gamma > 0$$

$$E_{\text{realization}} = E\gamma$$

$$M_{\text{op}}^c = 0$$

$$M_{\text{op}}[D^{c\delta}] = c \hat{t}^{\text{D}}$$

Proposed Photon Reading

A photon may be pictured as an internally timeless lightlike marker. Observed light travel may be read as D-flow moving relative to that marker.

$F_{\text{tu}}^{\text{Md}} = 0 \Rightarrow$ zero rest-mass persistence along the rail. Φ writes only at contact/closure: absorption, scattering, detection.

SR Reconciliation

$$\chi(v) = \frac{1}{\gamma} = d\frac{\tau}{dt} \text{ (} d\frac{\tau}{dt} \text{ is the SR identification, imported)}$$

$$\Delta t = \gamma \Delta \tau$$

$$L = L_0/\gamma$$

$$\chi(v) = d\frac{\tau}{dt} = c_C/c$$

$$c_C = c \cdot d\frac{\tau}{dt}$$

$$\text{photon : } \chi = 0 \Rightarrow c_C = 0 \text{ During -- to -- photon relative speed} = c$$

Typed Observed Readout

$$V_D = T(\Pi_{xy})$$

$$E_z = \text{rail -- normal energy channel}$$

$$F_{\text{tu}}^{\text{O}} \in V_D \times E_z$$

$$F_{\text{tu}}^{\text{O}} = \Theta_{\text{win}} \cdot \langle M_{\text{op}}[D^\delta], M_{\text{op}}^c \rangle$$

$$[F_{\text{tu}}^{\text{O}}] = \langle L \cdot T^{-1}, E \rangle$$

$$\text{massive readout} = \Theta_{\text{win}} \cdot \langle v_{\text{flow}}^{\text{D}}, M_{\text{op}}^c \rangle \text{ light readout} = \Theta_{\text{win}} \cdot \langle c \hat{t}^{\text{D}}, 0 \rangle$$

$$\text{geodesic} = \text{free -- fall readout}$$

$$\text{SR} = \text{relativistic readout}$$

PMM Statement

SR is recovered as the area-preserving, gauge-preserving deformation of the During rails.

M_{op} maps a During propagation mode to a velocity readout. M_{op}^c is an energy-valued rail-normal carrier participation readout. Massive carriers may have $|M_{\text{op}}^c| > 0$.

Photons have $M_{\text{op}}^c = 0$. The lightlike During readout is $M_{\text{op}}[D^{c\delta}] = c\hat{t}^{\text{D}}$. The photon-marker picture is a proposed PMM reading. Records are written at contact/closure.

1.7 6. FTU Information Storage

Information storage describes how D-phase inputs give rise to A-record probabilities and persistent Φ updates.

D-phase conditioning input

$$\diamond = (F_{\text{tu}}^y, F_{\text{tu}}^x | \delta, \Theta_{\text{win}}, \Theta_{\text{con}})$$

A-record alphabet

$$\Omega_a = \{\phi_1, \phi_2, \dots, \phi_i\}$$

A-transcription operator

$$\mathcal{T}_A = c^3 - \text{transcription}$$

c^3 transcription gives a probability law over A-records

$$\Theta_{\text{out}}[\diamond] = \mathcal{T}_A[\diamond] \rightarrow P(\Omega_a)$$

Closure Probability

$$p_i = |\langle \phi_i | \Psi \rangle|^2 = P(\phi_i | \diamond, \Theta_{\text{out}}[\diamond], \Theta_{\text{con}})$$

Closure Draw

$$K \sim \text{Cat}(\{p_i\})$$

Selected A-Record

$$\phi_K = \text{selected } A - \text{record}$$

Phase-memory update

$$\Phi_{n+1} = \lambda \Phi_n + (1 - \lambda) S_n - \Delta \tau \nabla \cdot (\Phi_n u_n)$$

A-Sector Persistence

$$\Phi' \in A$$

1.8 7. Phase Dynamics

This section gathers phase weights, memory writing, stress contrast, depth, regimes, and action terms.

PHASE PROGRESS

$$u = \text{local monotonic phase parameter } (B \rightarrow D \rightarrow A)$$

$$u_{\text{in}} < u_{\text{out}}$$

PHASE WEIGHTS

$$\mathcal{Z}_{\Theta} = (1 - \Theta_{\text{in}}[\Phi]) + \Theta_{\text{win}} + \Theta_{\text{out}}[\Diamond]$$

$$w_B = (1 - \Theta_{\text{in}}[\Phi])/\mathcal{Z}_{\Theta}$$

$$w_D = \Theta_{\text{win}}/\mathcal{Z}_{\Theta}$$

$$w_A = \Theta_{\text{out}}[\Diamond]/\mathcal{Z}_{\Theta}$$

$$w_B + w_D + w_A = 1$$

$$g_{\mu\nu}^{\text{D}}[\Phi, \delta] = g_{\mu\nu}^{\text{GR}}[\Phi] + h_{\mu\nu}^{\delta}[\delta] + h_{\mu\nu}^{\Phi\delta}[\Phi, \delta]$$

Gravitating potential sourced by phase memory

$$\nabla \cdot [\mu(|\nabla\psi|/a_0)\nabla\psi] = 4\pi G_N(\rho_b + \xi\Phi)$$

$$\mu(x) \approx 1 \text{ for } x \gg 1$$

$$\mu(x) \approx x \text{ for } x \ll 1$$

STRESS CONTRAST

$$\bar{T}_{\mu\nu}^{(B)} = \langle \hat{T}_{\mu\nu} \rangle_{\Psi_B} \quad \bar{T}_{\mu\nu}^{(D)} \approx T_{\mu\nu}^{(D)}$$

$$(\Delta T)^2 = (\bar{T}_{\mu\nu}^{(B)} - \bar{T}_{\mu\nu}^{(D)})(\bar{T}^{(B)\mu\nu} - \bar{T}^{(D)\mu\nu})$$

Action-contrast closure (dimensional form of the B–D mismatch)

$$\Delta S = S^{\text{D}} - S^{\text{B}} = mcd - \hbar$$

$$(\bar{T}^{(B)} - \bar{T}^{(D)}) = \Delta S / \hbar \text{ per FTU}$$

$$(\Delta T)_{\text{eff}}^2 = \left(\frac{mcd}{\hbar} - 1\right)^2 \cdot \left(1 - \frac{\Phi}{\Phi_{\text{max}}}\right)$$

On-shell vanishing

$$d = \frac{\hbar}{mc} \Rightarrow mcd = \hbar \Rightarrow \Delta S = 0 \Rightarrow (\Delta T)_{\text{eff}}^2 = 0$$

On-shell traversal writes no excess Φ from B-D action mismatch. The baseline A-record still closes through $\phi_K \rightarrow S_n \rightarrow \Phi_{n+1}$.

Off-shell departure supplies the excess Φ -writing source.

TRANSCRIPTION

$$\partial\Phi/\partial\tau_A + \nabla \cdot (\Phi u) = \Gamma_{\textcircled{a}}^{\Phi} \Theta_{\text{out}} (\Delta T)_{\text{eff}}^2 - \Phi/T_p$$

$$\Gamma_{\textcircled{a}}^{\Phi} = \Phi\text{-write coefficient} \quad \Gamma_{\textcircled{a}}^{\Phi} \sim k_B T$$

$[\Gamma_{\textcircled{a}}^{\Phi}]$ = action/time Φ is measured in action-record units in the transport law.

STRUCTURAL DEPTH

$$\delta[\Phi, \rho] = \ell_{\delta} \cdot |\nabla^2 \Phi| / (|\nabla \Phi| + \epsilon_{\Phi}) \cdot \frac{\rho}{(\rho + \rho_c)}$$

$$L_{\delta} = -(\frac{1}{2})\Theta_{\text{win}}(\partial_{\mu}\delta)(\partial^{\mu}\delta) - \frac{\alpha_{\delta}}{2}\delta^2 \cdot \frac{\rho_c}{(\rho + \rho_c)}$$

Horizon Acceleration Readouts

$$L_{\Lambda} = 1/\sqrt{\frac{\Lambda}{3}}$$

$$H_{\Lambda} = c/L_{\Lambda}$$

$$H_{\Lambda} = H_0 \sqrt{\Omega_{\Lambda}}$$

$$T_{\text{dS}} = \hbar c / (2\pi k_{\text{B}} L_{\Lambda})$$

$$\gamma_{\Lambda} = k_{\text{B}} T_{\text{dS}} / \hbar$$

$$\gamma_{\Lambda} = \frac{c}{(2\pi L_{\Lambda})}$$

$$\gamma_{\Lambda} = \frac{H_{\Lambda}}{2\pi}$$

$$a_{\Lambda} = c\gamma_{\Lambda}$$

$$a_{\Lambda} = cH_{\Lambda}/2\pi$$

Why a_0 Tracks cH_0

PMM does not treat $a_0 \approx cH_0$ as a loose coincidence. A horizon carries a thermal clock.

$$T_H = \hbar H / (2\pi k_{\text{B}})$$

clock gives an A-phase rate.

$$\gamma_H = k_{\text{B}} T_H / \hbar$$

$$\gamma_H = \frac{H}{2\pi}$$

C-gauge reads that rate as D-phase acceleration.

$$a_H = c\gamma_H$$

$$a_H = \frac{cH}{2\pi}$$

**The cosmic acceleration scale follows from horizon temperature: $\rightarrow$ A-phase clock
 $\rightarrow$ C-gauge acceleration readout**

For the present expansion horizon:

$$a_{H0} = cH_0/2\pi$$

Observed galaxy fits give:

$$a_{0, obs} \approx 1.16\text{--}1.20 \times 10^{-10} \frac{m}{s^2}$$

PMM read

So PMM explains the scale of a_0 through the horizon clock, while the remaining offset belongs to the empirical comparison between a_{H0} and $a_{0,obs}$.

C-Gauge Kinematic Readout

$$C \rightarrow c$$

$$a_\Lambda = c\gamma_\Lambda$$

$$a_\Lambda = cH_\Lambda/2\pi$$

$$a_\Lambda = c^2\sqrt{\frac{\Lambda}{3}}/2\pi$$

Present-Epoch Empirical Check

$$a_{H0} = cH_0/2\pi$$

$$a_{0,obs} \approx 1.16\text{--}1.20 \times 10^{-10} \frac{m}{s^2}$$

Horizon-Rate Fork

$$a_{H0} = cH_0/2\pi$$

$$a_{H0}/a_\Lambda = H_0/H_\Lambda$$

$$H_0/H_\Lambda = 1/\sqrt{\Omega_\Lambda}$$

$$H_0/H_\Lambda \approx 1.21$$

Test Readout

$$a_\Lambda \approx 8.63 \times 10^{-11} \frac{m}{s^2}$$

$$a_{H0} \approx 1.04 \times 10^{-10} \frac{m}{s^2}$$

$$a_{0,obs} \approx 1.16\text{--}1.20 \times 10^{-10} \frac{m}{s^2}$$

$$a_0, obs/a_\Lambda \approx 1.35\text{--}1.39$$

$$a_0, obs/a_{H0} \approx 1.12\text{--}1.15$$

PMM Acceleration Candidates

$$a_\Lambda = cH_\Lambda/2\pi$$

$$a_{H0} = cH_0/2\pi$$

$$a_{\text{PMM}, \Lambda} = a_\Lambda$$

$$a_{\text{PMM}, H_0} = a_{H0}$$

PMM Read

T_{ds} supplies the angular Λ -horizon rate $\gamma\Lambda$. *The C-gauge reads $\gamma\Lambda$ as the Λ -only D-phase acceleration a_Λ .* The full expansion horizon gives $a_{H0} = cH_0/2\pi$. The horizon-rate fork is a present-epoch conversion:

$a_{H0}/a_\Lambda = H_0/H_\Lambda = 1/\sqrt{\Omega_{\Lambda,0}}$. This is a horizon-rate comparison, not a Φ -retention law. The remaining empirical offset depends on which H_0 family and which galaxy-fit acceleration scale is compared.

REGIMES

$$|\nabla\psi| \gg a_0 \rightarrow \text{GR/Newtonian}$$

$$|\nabla\psi| \ll a_0 \rightarrow \text{MOND-like}$$

$$g \approx \sqrt{G_N \text{Ma}_0}/r v^4 \propto G_N \text{Ma}_0$$

FIELD ACTION

$$\mathcal{A} = \int d^4x \sqrt{-g_{\text{eff}}}$$

1.9 8. Standard Physics Mappings

This section shows how the PMM grammar maps into familiar physics and information models.

1.9.1 Shannon Information

The Shannon block treats A-record selection as a probability law and entropy bridge.

FTU Information Progression

Start with the After Axis

$$A = (c^3)_{k_B}$$

Depth rule

$$\delta \notin B$$

$$\delta \in D$$

$$\delta \notin A$$

D-phase conditioning input

$$\diamond = (F_{\text{tu}}^y, F_{\text{tu}}^x \mid \delta, \Theta_{\text{win}}, \Theta_{\text{con}})$$

A-record alphabet

$$\Omega_a = \{\phi_1, \phi_2, \dots, \phi_i\}$$

c^3 transcription gives a probability law over A-records

$$\Theta_{\text{out}}[\diamond] \equiv \mathcal{T}_A[\diamond] \rightarrow P(\Omega_a)$$

Contact shapes closure probability

$$\Theta_{\text{con}} \rightarrow \{p_i\}$$

Closure probability

$$p_i = P(\phi_i \mid \diamond, \Theta_{\text{out}}[\diamond], \Theta_{\text{con}})$$

Normalization

$$\Sigma_i p_i = 1$$

Closure draw

$$K \sim \text{Cat}(p_1, p_2, \dots, p_i)$$

Selected A-record

$$\phi_K = \text{selected } A - \text{record}$$

Phase-memory update

$$\Phi_{n+1} = \lambda \Phi_n + (1 - \lambda) S_n - \Delta \tau \nabla \cdot (\Phi_n u_n)$$

Persistence

$$\Phi_{n+1} \in A$$

Single-record information

$$I(\phi_i) = -\log_2(p_i)$$

FTU-Shannon information

$$H_{(FTU)}(\diamond) = -\sum_i p_i \log_2(p_i)$$

Boltzmann bridge

$$S_{(FTU)} = k_B \ln(2) H_{(FTU)}$$

Ultra compact

$$\mathcal{T}_A[\diamond] \rightarrow P(\Omega_a)$$

$$p_i = P(\phi_i \mid \diamond, \Theta_{\text{out}}[\diamond], \Theta_{\text{con}}) \, K \sim \text{Cat}(\{p_i\})$$

$$\Phi_{n+1} = \lambda \Phi_n + (1 - \lambda) S_n - \Delta \tau \nabla \cdot (\Phi_n u_n)$$

$$\Phi_{n+1} \in A$$

$$H_{(FTU)} = -\sum_i p_i \log_2(p_i)$$

$$S_{(FTU)} = k_B \ln(2) H_{(FTU)}$$

1.9.2 Double Slit Experiment

The double slit block reads interference as open D traversal followed by a single A-record closure.

definitions

$$C_1 = \text{entry through the slit geometry}$$

$$C_2 = \text{traversal across the open path field}$$

$$C_3 = \text{screen} - \text{level closure}$$

$$@ = \text{contact event that opens the traversal}$$

$$\Phi = \text{the written detector record (the } \textit{dot} \text{ on the screen)}$$

Proposed PMM reading: lightlike D-flow traverses the slit field

relative to the photon marker

@ contact opens the traversal C_1 admits both slit possibilities C_2 carries the interference weighting C_3 closes at the screen Φ records one dot (the selected A-record ϕ_K)

1.9.3 Heisenberg

The Heisenberg block introduces uncertainty into the B-to-D entry and the orientation spread.

Start with Before Axis

$$B = (c^0)_{\text{mp}}$$

B-position scale

$$F_{\text{tu}}^{\text{xb}} = \frac{\hbar}{(m_p c)}$$

B-time scale

$$F_{\text{tu}}^{\text{Tb}} = \frac{\hbar}{(m_p c^2)}$$

B-energy scale

$$F_{\text{tu}}^{\text{Eb}} = m_p c^2$$

From energy, get the B-momentum scale:

$$p^b = E^b/c = m_p c$$

Multiply the B-position scale by the B-momentum scale:

$$F_{\text{tu}}^{\text{xb}} \cdot p^b = [\frac{\hbar}{(m_p c)}] \cdot [m_p c] = \hbar$$

Quantum spread rule:

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

Uncertainty-Extended During Gate

$$C_2[\Phi, \delta, \Sigma] = \Theta_{\text{win}}$$

$$\Sigma \supseteq \Delta x \Delta p, \text{ the } B - \text{ phase instance}$$

1.9.4 Einstein General Relativity

The GR block treats spacetime geometry as the depth-free D-metric limit.

Starting from the During Axis where spacetime exists.

$$D = (c^1, c^2)_C$$

Rails

$$c^1 = \text{During traversal / time} - \text{gravity role}$$

$$c^2 = \text{relational realization / energy} - \text{mass role}$$

$$C = \text{light} - \text{speed causal gauge}$$

D-Phase Metric

$$D \rightarrow g_{\mu\nu}^D$$

PMM Metric Form

$$g_{\mu\nu}^D[\Phi, \delta] = g_{\mu\nu}^{\text{GR}}[\Phi] + h_{\mu\nu}^\delta[\delta] + h_{\mu\nu}^{\Phi\delta}[\Phi, \delta]$$

GR Sector

$$\delta \rightarrow 0$$

Depth-Free GR Limit

$$g_{\mu\nu}^D(\Phi, 0) = g_{\mu\nu}^{\text{GR}}(\Phi)$$

Curvature From D-Metric

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}$$

Energy-Mass Source

$$T_{\mu\nu}$$

Einstein Field Equation

$$G_{\mu\nu} = (8\pi G_N/c^4)T_{\mu\nu}$$

D-Geodesic

$$\frac{d^2 x^\mu}{d(\tau^D)^2} + \Gamma_{\alpha\beta}^\mu[g^D] \cdot \left(\frac{dx^\alpha}{d\tau^D} \right) \left(\frac{dx^\beta}{d\tau^D} \right) = 0$$

One Line Bridge

$$D = (c^1, c^2)_C \rightarrow g_{\mu\nu}^D \rightarrow G_{\mu\nu} = (8\pi G_N/c^4)T_{\mu\nu}$$

1.9.5 Special Relativity: Time Dilation and Length Contraction

This block derives the special-relativistic readout from the During rail budget. In PMM language, relativity occurs inside D because D is where traversal, realized length, and the light-speed gauge are active.

Start with During

$$D = (c^1, c^2)_C$$

Rail assignments

$$c^1 = \text{During traversal/time role}$$

$$c^2 = \text{relational realization/length role}$$

C = light – speed causal gauge

The relational origin of c^2 is defined in Part II. SR deformation acts on the established local pair:
 $D = (c^1, c^2)_C$.

Velocity readout

massive:

$$v_{xy} = d(F_{tu}^{xd}, F_{tu}^{yd})/dt$$

$$v = |v_{xy}| 0 \leq v < c$$

Light limit

$$v = c$$

PMM rail-budget derivation of χ

$$D = (c^1, c^2)_C$$

$$v = \text{external } D - \text{readout speed}$$

$$c_C = \text{speed} - \text{valued carrier participation readout}$$

$$[v] = [c_C] = L \cdot T^{-1} = m \cdot s^{-1}$$

$$C = \text{total } D \text{ causal gauge}$$

Gauge budget

$$c^2 = v^2 + c_C^2$$

Define χ as carrier participation fraction

$$c_C = c\chi$$

Derive χ

$$c^2 = v^2 + c^2\chi^2$$

$$1 = \frac{v^2}{c^2} + \chi^2$$

$$\chi^2 = 1 - \frac{v^2}{c^2}$$

$$\beta_v = \frac{v}{c}$$

$$\chi(v) = \sqrt{1 - \beta_v^2}$$

$$\gamma = \frac{1}{\chi}$$

Relativistic depth-modified During

$$D^{\gamma\delta} = (\frac{c^1}{\chi}, c^2\chi)_{\mathbf{C}^\delta}$$

$$D^{\gamma\delta} = (\gamma c^1, \frac{c^2}{\gamma})_{\mathbf{C}^\delta}$$

Meaning

c^1 expands by γ c^2 contracts by γ $\mathbf{C}^\delta$ carries the depth-modified D-gauge χ controls carrier participation in During

Time dilation map

$$F_{\text{tu}}^{\text{Td}}(0) = \frac{d}{c} F_{\text{tu}}^{\text{Td}}(v) = F_{\text{tu}}^{\text{Td}} \frac{(0)}{\chi} F_{\text{tu}}^{\text{Td}}(v) = \gamma \cdot F_{\text{tu}}^{\text{Td}}(0) F_{\text{tu}}^{\text{Td}}(v) = \gamma d/c$$

Standard SR form

$$\Delta t = \gamma \Delta \tau$$

PMM read

Moving traversal takes longer in the external D-readout because the c^1 rail expands by γ .

Length contraction map

$$F_{\text{tu}}^{\text{yd}}(0) = L_0 F_{\text{tu}}^{\text{yd}}(v) = F_{\text{tu}}^{\text{yd}}(0) \chi F_{\text{tu}}^{\text{yd}}(v) = F_{\text{tu}}^{\text{yd}} \frac{(0)}{\gamma} F_{\text{tu}}^{\text{yd}}(v) = L_0/\gamma$$

Elementary on-shell length

$$L_0 = \frac{\hbar}{(mc)}$$

Contracted D-readout

$$F_{\text{tu}}^{\text{yd}}(v) = \frac{\hbar}{(\gamma mc)}$$

Standard SR form

$$L = L_0/\gamma$$

PMM read

Moving the realized length shortens the external D-readout because the c^2 rail contracts by γ .

One Line Bridge

$$D^\delta = (c^1, c^2)_{\mathcal{C}^\delta} \rightarrow$$

$$c^2 = v^2 + c_C^2 \rightarrow c_C = c\chi \rightarrow \beta_v = \frac{v}{c} \rightarrow \chi^2 = 1 - \beta_v^2 \rightarrow \chi(v) = \sqrt{1 - \beta_v^2}$$

$$\rightarrow \gamma = \frac{1}{\chi} \rightarrow D^{\gamma\delta} = (\frac{c^1}{\chi}, c^2\chi)_{\mathcal{C}^\delta} = (\gamma c^1, \frac{c^2}{\gamma})_{\mathcal{C}^\delta} \rightarrow \Delta t = \gamma\Delta\tau \text{ and } L = L_0/\gamma$$

Light Limit

$$v = c \Rightarrow \chi = 0 \Rightarrow c_C = 0$$

$$M_{\text{op}}^c = 0$$

$$M_{\text{op}}[D^{c\delta}] = c\hat{t}^{\mathcal{D}}$$

Proposed Photon Reading

photon = internally timeless lightlike marker observed light-speed readout = D-flow relative to marker

1.9.6 Schwarzschild

Schwarzschild = the depth-free, spherical, vacuum D-metric around a central mass source.

Start with During

$$D = (c^1, c^2)_{\mathcal{C}}$$

Depth-free GR sector

$$\delta \rightarrow 0$$

$$g_{\mu\nu}^{\text{D}}(\Phi, 0) = g_{\mu\nu}^{\text{GR}}(\Phi)$$

Vacuum exterior

$$T_{\mu\nu} = 0 \text{ for } r > R$$

Spherical central source

$$\Phi \rightarrow \Phi_M(r)$$

Einstein vacuum equation

$$G_{\mu\nu} = 0$$

Schwarzschild metric

$$ds^2 = -(1 - 2G_N \frac{M}{c^2 r})c^2 dt^2 + (1 - 2G_N \frac{M}{c^2 r})^{-1} dr^2 + r^2 d\Omega^2$$

Schwarzschild radius

$$r_s = 2G_N \frac{M}{c^2}$$

One Line Bridge

$$D = (c^1, c^2)_{\text{C}} \rightarrow \delta \rightarrow 0 \rightarrow g_{\mu\nu}^{\text{D}}(\Phi, 0) = g_{\mu\nu}^{\text{GR}}(M, r) \rightarrow \text{Schwarzschild}$$

1.10 9. Tests

The following is a compact list of tests performed on PMM.

1.10.1 Depth Regulation Test

Purpose: test whether δ can regularize extreme Φ compression while preserving the GR limit.

Test Metric

$$g_{\mu\nu}^{\text{D}}[\Phi, \delta] = g_{\mu\nu}^{\text{GR}}[\Phi] + h_{\mu\nu}^{\delta}[\delta] + h_{\mu\nu}^{\Phi\delta}[\Phi, \delta]$$

GR Recovery

$$\delta \rightarrow 0$$

$$h_{\mu\nu}^{\delta}[\delta] \rightarrow 0 \, h^{\Phi\delta}{}_{\mu\nu}[\Phi, \delta] \rightarrow 0 \, g^{\text{D}}{}_{\mu\nu}[\Phi, \delta] \rightarrow g_{\mu\nu}^{\text{GR}}[\Phi]$$

Depth Activation

$$\rho_c \approx \rho_P$$

$$\rho \ll \rho_c \Rightarrow \delta \rightarrow 0$$

$$\rho \gtrsim \rho_c \Rightarrow \delta \rightarrow \delta_{\text{sat}}$$

Core Scale

$$r_{\text{core}} = \ell_{\delta} \delta_{\text{sat}}$$

Core Regularization

$$r_D^2 \rightarrow r_D^2 + r_{\text{core}}^2$$

Toy Radial Lapse

$$A_{\text{PMM}}(r) = 1 - 2G_N M / (c^2 \sqrt{r_D^2 + r_{\text{core}}^2}) + \beta \delta(r)$$

Exterior Limit

$$\delta \rightarrow 0 \quad r_D \gg r_{\text{core}}$$

$$A_{\text{PMM}}(r) \rightarrow 1 - 2G_N \frac{M}{(c^2 r_D)}$$

Core Limit

$$r_D \rightarrow 0 \, A_{\text{PMM}}(0) \text{ finite}$$

Repulsive Depth Test

$$\beta > 0$$

Pass Condition

δ activates near ρ_P δ saturates at δ_{sat} $A_{\text{PMM}}(0)$ finite $A_{\text{PMM}}(r)$ returns to GR outside the core

$$F_{\text{tu}}^s \geq \hbar^3$$

$$N \geq 1$$

PMM Interpretation

A black-hole remnant is a minimum nonzero closure record stabilized by saturated D-phase depth.

1.11 10. Extensions

This section adds optional orientation behavior while preserving the core $B \circ D \circ A$ path.

1.11.1 Spinor Extension

Spinorial behavior is carried as a two-rail phase field inside the D-phase depth channel. The spinor influences contact and closure selection. At closure, the selected spin-conditioned record is written into A through Φ .

D-Phase Spinor Rail Field

$$\Sigma^D = (\Sigma_1, \Sigma_2)^T \Sigma_1 \leftrightarrow c^1 \Sigma_2 \leftrightarrow c^2$$

Spinor Symmetry (postulated)

$$\Sigma^D(2\pi) = -\Sigma^D(0)$$

$$\Sigma^D(4\pi) = \Sigma^D(0)$$

Proposed PMM Two-Rail Phase Reading

A 2π rotation reverses both rail phases. A 4π rotation restores the original two-rail state. The 2π phase reversal becomes detectable through interference with an unrotated path.

Spinor-Extended Contact

$$\Theta_{\text{con}}^{(\Sigma)} = \{C_1[@, \Phi], C_2[\Phi, \delta, \Sigma^D], C_3[\Phi, \Sigma^D]\}$$

Spinor-Extended During

$$D^\delta \Sigma = ((c^1, c^2)_{C^\delta}, \Sigma^D)$$

Interpretation

Σ^D carries spinor phase across the c^1 and c^2 rails. Spinor phase is carried through the D-phase depth channel under the C-gauge.

The $B \rightarrow D \rightarrow A$ path stays unchanged.

1.11.2 Electromagnetic Extension

Electromagnetism enters PMM as a D-phase-field contribution to the local stress-energy content, written into Φ .

EM Phase Contributions

$\text{EM}_B \rightarrow$ possible interaction

$\text{EM}_D \rightarrow$ active field readout

$\text{EM}_A \rightarrow$ recorded contribution to Φ

EM D-Phase Contribution

$$D = (c^1, c^2)_C$$

$$T_{\mu\nu}^{\text{total}}$$

$$T_{\mu\nu}^{\text{mass}}$$

$$T_{\mu\nu}^{\text{EM}}$$

$$T_{\mu\nu}^{\text{other}}$$

Φ Write

$T_{\mu\nu}^{\text{EM}}$ contributes to the D – phase mismatch encoded into Φ .

$$\Phi = \Phi[T_{\mu\nu}^{\text{total}}, \Theta_{\text{win}}, \Theta_{\text{con}}]$$

Gravity Coupling

$$g_{\mu\nu}^{\rm D}[\Phi,\delta]$$

$$g_{\mu\nu}^{\rm GR}[\Phi]$$

$$h_{\mu\nu}^{\delta}[\delta]$$

$$h_{\mu\nu}^{\Phi\delta}[\Phi,\delta]$$

Depth Coupling

$$\delta[\Phi,\rho]$$

$$\ell_\delta \cdot |\nabla^2 \Phi|/(|\nabla \Phi| + \epsilon_\Phi) \cdot \frac{\rho}{(\rho + \rho_c)}$$

D-to-A Light Readout

Proposed PMM reading:

Lightlike D-flow carries the source-to-observer readout relative to the photon marker.

$$D^\delta \,=\, (c^1,c^2)_{\mathcal{C}^\delta}$$

$$z \,=\, c^1 \times_r c^2$$

$$\rightarrow (x,y;\,z_{\rm dir},z_{\rm mag})$$

$$A \,=\, \Phi(x,y,T)$$

z is rail-normal During readout

Perceived Depth Compression

far distance

→ smaller angular spread

→ weaker parallax

→ reduced *z cue*

→ compressed perceived depth

Interpretation

Electromagnetism carries the D-phase readout into the observer's A-side record. Depth is active in D, while distance compresses usable z information into a thinner x,y readout.

2 PART II — MULTI-FTU VANTAGE

This part introduces the second FTU and derives the relations that require a partner: relational realization of c^2 , contact and closure accounting, the origin of the phase-memory/depth inputs used in Part I, the gravitational crease and bond, and the aggregate recoveries built on them.

2.1 11. Relational Realization

FTU_{*i*} contributes c_i^1

FTU_{*j*} contributes c_j^1

@_{*ij*} : (c_i^1, c_j^1) → paired per-vantage realization

$$\mathcal{R}_i(c_j^1 \mid c_i^1) = c_{j \rightarrow i}^2$$

$$\mathcal{R}_j(c_i^1 \mid c_j^1) = c_{i \rightarrow j}^2$$

$\mathcal{R}_i$ is the defining per-vantage realization map. It carries the partner traversal contribution into the realization role of the receiving FTU. $\mathcal{R}$ adds no independent angular rotation or free scale. The receiving FTU supplies the c^2 scale $L_{0i} = \hbar/(m c)$; partner orientation information is carried into the relational readout.

Geometric-order bridge

Two coupled c^1 -order traversal contributions establish a c^2 -order relational realization. $\mathcal{R}_i$ supplies the defining per-vantage operation.

Relational index convention

When multiple FTU vantages appear together, *i* labels the receiving vantage and *j* the partner/source. Local quantities remain unindexed unless the comparison itself requires explicit *i/j* labels.

Contact Closure Counting

@_{*ij*} = one physical contact event

$$N(@_{ij}) = 1$$

$$A_i[@_{ij}] = \text{vantage-i record of that closure}$$

$$A_j[@_{ij}] = \text{vantage-j record of that closure}$$

A_i and A_j are two relational records of one closure event. $F_{\text{tu}}^{s(i)}$ and $F_{\text{tu}}^{s(j)}$ are per-vantage action readouts of that same closure and are never added as two closures.

$$N_{\text{total}} = \sum_a N(@_a) \Rightarrow \text{closures count per physical contact event}$$

2.2 12. Contact Dynamics and Closure

This block defines each core operator's role before the geometry and information sections use it.

Wave function collapse

$$@ = \text{contact event that opens FTU traversal}$$

Contact operator

$$\Theta_{\text{con}} = \text{Contact} \mid \text{the Observer}$$

$$\Theta_{\text{con}} = \{C_1[@, \Phi], C_2[\Phi, \delta, \Sigma], C_3[\Phi, \Sigma]\}$$

Contact constraints

$$C_1[@, \Phi] = \Theta_{\text{in}}[\Phi]$$

$$C_2[\Phi, \delta, \Sigma] = \Theta_{\text{win}}$$

$$C_3[\Phi, \Sigma] = \Theta_{\text{out}}[\diamond]$$

Constraint roles

$$C_1 = \text{initiator constraint}$$

$$C_2 = \text{corrector constraint}$$

$$C_3 = \text{selector constraint}$$

Phase-memory field

$$\Phi = \text{persistent record field}$$

$$\Phi_{n+1} = \lambda \Phi_n + (1 - \lambda) S_n - \Delta \tau \nabla \cdot (\Phi_n u_n)$$

Persistence

$$\Phi_{n+1} \in A$$

Depth rule

$$\delta \notin B$$

$$\delta \in D$$

$$\delta \notin A$$

Structural Depth Field

L_δ is the dynamical *law* for δ . The algebraic form below is *its*

static (equilibrium) solution : the value δ relaxes to as $\partial \delta / \partial t \rightarrow 0$

(slow / quasi-static limit).

$$L_\delta = -(\frac{1}{2})\Theta_{\text{win}}(\partial_\mu \delta)(\partial^\mu \delta) - \frac{\alpha_\delta}{2}\delta^2 \cdot \frac{\rho_c}{(\rho + \rho_c)}$$

$$\delta[\Phi,\rho] = \ell_\delta \cdot |\nabla^2 \Phi|/(|\nabla \Phi| + \epsilon_\Phi) \cdot \frac{\rho}{(\rho + \rho_c)}$$

Orientation field

$$\Sigma = \text{closure spread } \Sigma \supseteq \Delta x \Delta p, \text{ the } B - \text{phase instance}$$

FTU Contact Dynamics

$$F_{\text{tu}}^{\textcircled{a}} = \Theta_{\text{con}} B[\textcircled{a}, \Phi] \rightarrow \Theta_{\text{in}}[\Phi] \rightarrow D(\Theta_{\text{win}}) \rightarrow \Theta_{\text{out}}[\diamond] \rightarrow A(\Phi')$$

Contact primitive (what @ IS)

$$@ : (F_{\text{tu},1}, F_{\text{tu},2}) \rightarrow \text{one physical contact bond}$$

$$q = \Delta p_{\text{xy}}$$

z_{dir} is the gate-derived orientation readout. Per-vantage depth is generated after relational realization.

Per-vantage FTUs (one contact, two perspectives)

Each interacting FTU contributes a c^1 -order During traversal to contact.

$$\text{FTU}_1 \text{ contributes } c_1^1 \text{ FTU}_2 \text{ contributes } c_2^1$$

$$@_{12}$$

$$: (c_1^1, c_2^1) \rightarrow \text{paired per - vantage realization}$$

Per-vantage relational realization

$$\mathcal{R}_1(c_2^1 \mid c_1^1) = c_{2 \rightarrow 1}^2$$

$$\mathcal{R}_2(c_1^1 \mid c_2^1) = c_{1 \rightarrow 2}^2$$

$\mathcal{R}_1$ and $\mathcal{R}_2$ are per-vantage maps. Each carries partner orientation information into the receiving FTU's realization role, adds no independent angular rotation or free scale, and uses the receiving FTU's local realization span.

$$L_{01} = \frac{\hbar}{(m_1 c)}$$

$$L_{02} = \frac{\hbar}{(m_2 c)}$$

Per-vantage During

$$D_1 = (c_1^1, c_{2 \rightarrow 1}^2)_C$$

$$D_2 = (c_2^1, c_{1 \rightarrow 2}^2)_C$$

Roles

c^1 = self traversal contribution

c^2 = partner traversal relationally realized through coupling

Depth readout

$$z_1 = c_1^1 \times_r c_{2 \rightarrow 1}^2$$

$$z_2 = c_2^1 \times_r c_{1 \rightarrow 2}^2$$

$$z_{\text{mag}}^{(1)} = d_1 L_{01} \sin(\theta_G)$$

$$z_{\text{mag}}^{(2)} = d_2 L_{02} \sin(\theta_G)$$

Each vantage processes the same contact through its own Θ_{con}

$$C_1[@, \Phi]$$

$$C_3[\Phi, \Sigma] = \Theta_{\text{out}}[\diamond]$$

Contact = two vantage-FTUs sharing one @

$$@ (F_{\text{tu},1}, F_{\text{tu},2}) \Rightarrow \{\Theta_{\text{con}}^{(1)}, \Theta_{\text{con}}^{(2)}\}$$

$$\Theta_{\text{con}}^{(1)} = \{C_1[@, \Phi_1], C_2[\Phi_1, \delta, \Sigma], C_3[\Phi_1, \Sigma]\}$$

$$\Theta_{\text{con}}^{(2)} = \{C_1[@, \Phi_2], C_2[\Phi_2, \delta, \Sigma], C_3[\Phi_2, \Sigma]\}$$

Common contact quantities: $@_{12}$, $|q|$. Vantage-local depth readouts: z_1 , z_2 .

$$z_{\text{dir}}^{(1)} = -z_{\text{dir}}^{(2)}$$

Opposite sign per vantage; the bond has two ends.

$$q_1 + q_2 = -q_{\text{carrier}}$$

Conservation across the two vantages.

Multi-FTU Bond Readout

$$\text{bond}_{ij} = \text{Lock}(@_{ij}, \Theta_{\text{con}}^{(i)}, \Theta_{\text{con}}^{(j)})$$

$$\kappa_{ij} = |\text{bond}_{ij}|$$

Universal Gravitational Crease

G = universal FTU – to – FTU gravitational crease

θ_G = angular measure of the G crease

$$\theta_G = 45^\circ$$

$$\tan(\theta_G) = 1$$

$$\sin(\theta_G) = 1/\sqrt{2}$$

θ_G measures the G crease through the coupled c^1/c^2 D-phase rail geometry created at FTU-to-FTU contact. $\tan(\theta_G)$ governs the relational gravity projection. $\sin(\theta_G)$ governs the rail-normal depth projection.

$\hat{n}_{ij}$ = inward relational unit direction

Pairwise Gravity Tug

$$T_{u\,ij}^G = \tan(\theta_G)\kappa_{ij}\hat{n}_{ij}$$

Shorthand

$$T_u^G \equiv T_{u\,ij}^G$$

Realized Gravity-Tug Rule

$$\kappa_{ij} > 0 \Rightarrow T_{u\,ij}^G \neq 0$$

Meaning

The FTU carries the intrinsic time gradient $F_{\text{tu}}^G = \nabla(F_{\text{tu}}^T)$. θ^G gives the gradient its universal relational crease projection. Contact supplies the relational bond. Bond density supplies local magnitude. The bond-density gradient supplies inward direction.

Contact-Bond Area Rule

K_j = total scalar bond supply of source j

$\kappa_{ij}(r)$ = local contact – bond density available to i from j

$$\oint \kappa_{ij}(r) dA = K_j$$

$$A(r) = 4\pi r^2$$

$$\kappa_{ij}(r) = K_j/4\pi r^2$$

Action-Cube Bond Quantization (one closure = one $\hbar$ of bond)

$$K_j \cdot L_{0j} = S_j^D|_{\text{on-shell}} = \hbar$$

$$L_{0j} = \frac{\hbar}{(m_j c)}$$

$$\Rightarrow K_j = m_j c$$

$\Rightarrow \kappa_0 = c$ (action-cube bond scale; forced by cube constants, not fitted)

Gravitational Bond Scale (distinct from the action scale)

$\kappa_0 = c$ sources the cube, NOT gravity. Gravity needs Newton's units.

$$\kappa_G = 4\pi\hbar c/m_p^2 (= 4\pi G_N; \text{magnitude set by the Planck mass})$$

$$\Rightarrow K_j = \kappa_G m_j$$

$$\Rightarrow \kappa_{ij}(r) = \kappa_G m_j/4\pi r^2 = G_N m_j/r^2 \text{ (} 4\pi \text{ cancels)}$$

Angular Gravity Bridge

$\theta_G = 45^\circ \Rightarrow \tan(\theta_G) = 1$. $\tan(\theta_G)$ supplies the relational gravity projection; magnitude lives in κ_G

$$T_{u ij}^G = \tan(\theta_G) \kappa_{ij} \hat{n}_{ij}$$

$$\kappa_{ij}(r) = \kappa_G m_j/4\pi r^2 = G_N m_j/r^2$$

$$\Rightarrow T_{u ij}^G = G_N m_j \cdot \hat{n}_{ij}/r^2$$

$$\text{If } K_j = \kappa_G m_j :$$

$$\kappa_{ij}(r) = G_N m_j / r^2$$

2.2.1 Contact and Information Readout

This section links contact, closure probability, A-record selection, and Φ update.

2.2.2 Contact Probability Dynamics

Contact dynamics describes how Θ_{con} guides entry, traversal correction, and record selection

Wave function collapse

$$\text{Wave collapse} = @$$

$$C_1 = \text{initiator constraint}$$

$$C_2 = \text{corrector constraint}$$

$$C_3 = \text{selector constraint}$$

Contact

$$\Theta_{\text{con}} = \text{Contact} \mid \text{the Observer}$$

$$\Theta_{\text{con}} = \{C_1[@, \Phi], C_2[\Phi, \delta, \Sigma], C_3[\Phi, \Sigma]\}$$

$$C_1[@, \Phi] = \Theta_{\text{in}}[\Phi]$$

$$C_2[\Phi, \delta, \Sigma] = \Theta_{\text{win}}$$

$$C_3[\Phi, \Sigma] = \Theta_{\text{out}}[\diamond]$$

$$\Sigma = \text{closure spread}; \Delta x \Delta p \text{ is the } B - \text{phase instance}$$

D-phase entry is handled by $\Theta_{\text{in}}[\Phi]$

Contact shapes closure probability

$$\Theta_{\text{con}} \rightarrow \{p_i\}$$

2.3 13. Gravity

Gravity begins inside each FTU as the time gradient $F_{\text{tu}}^{\text{G}} = \nabla(F_{\text{tu}}^{\text{T}})$.

Through contact, that gradient gains relational expression as an inward bond response.

G is the universal gravitational crease governing the boundary relationship between interacting FTUs; θ^{G} is its angular measure in their coupled D-phase c^1/c^2 rail geometry. Its tan projection enters relational gravity, while $\kappa_G = 4\pi G^{\text{N}}$ supplies its measurable strength.

Bond density sets the local response magnitude, and $\nabla\kappa$ identifies the source-ward direction. Φ records completed bond history.

Starting with D-Phase rail structure

$$D = (c^1, c^2)_C$$

Gravity

Every FTU carries the intrinsic time-gradient readout. The universal gravitational crease G governs the boundary relationship between interacting FTUs.

$$G = \text{universal FTU} - \text{to} - \text{FTU gravitational crease}$$

$$\theta_G = \text{angular measure of } G$$

$$\theta_G = 45^\circ$$

Intrinsic and Relational Gravity Rule

Intrinsic: Every FTU carries the time-gradient readout

$$F_{\text{tu}}^{\text{G}} = \nabla(F_{\text{tu}}^{\text{T}}).$$

The inter-FTU boundary relation is governed by G. θ_G supplies its angular measure.

Relational: A measurable gravitational tug requires a bond between FTUs.

Expression:

$$T_{u\text{ij}}^{\text{G}} = \tan(\theta_G)\kappa_{\text{ij}}\hat{n}_{ij}$$

Meaning: Gravity originates as the intrinsic FTU time gradient. θ_G gives that gradient its relational crease projection. Contact allows the gradient to act between FTUs.

Gravity becomes relationally expressed through multi-FTU contact.

T_u^G carries the pairwise directional bond response. G^D carries the aggregate multi-bond gravity readout. Φ records completed bond history. δ modulates the D-phase geometry.

Definitions

g^D = full D – phase metric geometry

$g_{\mu\nu}^D[\Phi, \delta]$ = full D – phase metric generated by Φ and modulated by δ

$g_{\mu\nu}^{\text{GR}}[\Phi]$ = GR – limit metric recovered when $\delta \rightarrow 0$

$h_{\mu\nu}^\delta[\delta]$ = depth – only correction to the D – phase metric

$h_{\mu\nu}^{\Phi\delta}[\Phi, \delta]$ = memory – depth coupling correction to the D – phase metric

$\delta(r)$ = local depth response generated by Φ and density ρ

δ_{sat} = finite saturated depth value reached near the Planck-density core

r_D = D – phase radial compression coordinate

$A_{\text{PMM}}(r)$ = *toy* radial lapse used to test core regularization

β = positive coupling strength of the repulsive depth correction

$r_{\text{core}} = \ell_\delta \cdot \delta_{\text{sat}}$ = core regularization scale

Gravity Readout

$$F_{\text{tu}}^{\text{Gd}} = \nabla\left(\frac{d}{c}\right)$$

Multi-FTU Gravity Readout (G^D as theorem, not primitive)

G^D is the multi – bond aggregate of the single primitive $F_{\text{tu}}^G = \nabla(F_{\text{tu}}^T)$,

diluted by the Area Rule. It is a consequence, not a second primitive.

$$G_{ij}^D \equiv T_{u\,ij}^G$$

$$T_{u\,ij}^G = \tan(\theta_G) \kappa_{ij}(r) \hat{n}_{ij}$$

$$\kappa_{ij}(r) = K_j / 4\pi r_{ij}^2$$

$$G_i^D = \Sigma_j T_{u\,ij}^G$$

$$G_i^D = \Sigma_j \tan(\theta_G) K_j \hat{n}_{ij} / 4\pi r_{ij}^2$$

$$\text{If } K_j = \kappa_G m_j = 4\pi G_N m_j :$$

$$G_i^D = \Sigma_j \tan(\theta_G) \cdot G_N m_j \cdot \hat{n}_{ij} / r_{ij}^2 = \Sigma_j G_N m_j \hat{n}_{ij} / r_{ij}^2 (\tan(\theta_G) = 1)$$

Primitive Reduction

$$\theta_G = 45^\circ, \tan(\theta_G) = 1 \text{ (relational gravity projection)}$$

Magnitude carried by $\kappa_G = 4\pi \hbar c / m_p^2$, the Planck-mass gravity scale.

The action bridge $(\frac{c^3}{\hbar})$ does NOT close : units $m \cdot kg^{-1} \cdot s^{-2}$,

short of Newton's $m^3 \cdot kg^{-1} \cdot s^{-2}$ by m^2 . Correct bridge is $\hbar c / m_p^2 = G_N$.

$$G_i^D = \Sigma_j G_N m_j \hat{n}_{ij} / r_{ij}^2$$

$$\hat{n}_{ij} = -\nabla \kappa / |\nabla \kappa| \text{ emerges inward : bonds concentrate where}$$

closures fire (the source).

Source-ward direction is derived from where the

record is written, not posited.

$\tan(\theta^G) = 1$ supplies the relational gravity projection; magnitude is supplied by κ_G .

$$\kappa_G = 4\pi G^N \text{ supplies the gravitational magnitude scale.}$$

κ_{ij} supplies local bond density; $\hat{n}_{ij}$ supplies relational direction.

Φ records scalar bond history after closure. G_i^D carries the inward field readout.

Depth enabled During

$$D^\delta = (c^1, c^2)_{\mathbf{C}^\delta}$$

$$z = c^1 \times_r c^2$$

$$D^\delta \rightarrow (x, y; z_{\text{dir}}, z_{\text{mag}})$$

$$D^\delta \rightarrow x_D^\mu = (ct, x, y)$$

$$D^\delta \rightarrow ds_D^2 = g_{\mu\nu}^{\text{D}}[\Phi, \delta] dx^\mu dx^\nu$$

$$\text{Gravity} \rightarrow g_{\mu\nu}^{\text{D}}[\Phi, \delta]$$

Metric Decomposition

$$g_{\mu\nu}^{\text{D}}[\Phi, \delta] = g_{\mu\nu}^{\text{GR}}[\Phi] + h_{\mu\nu}^\delta[\delta] + h_{\mu\nu}^{\Phi\delta}[\Phi, \delta]$$

GR Limit

$$\delta \rightarrow 0$$

$$h_{\mu\nu}^\delta[\delta] \rightarrow 0 \quad h^{\Phi\delta}_{\mu\nu}[\Phi, \delta] \rightarrow 0 \quad g^{\text{D}}_{\mu\nu}[\Phi, \delta] \rightarrow g^{\text{GR}}_{\mu\nu}[\Phi]$$

Planck-Core Depth Regularization

$$r_{\text{core}} = \ell_\delta \cdot \delta_{\text{sat}}$$

$$r_D^2 \rightarrow r_D^2 + r_{\text{core}}^2$$

Toy Radial Lapse

$$A_{\text{PMM}}(r) = 1 - 2G_N M / (c^2 \sqrt{r_D^2 + r_{\text{core}}^2}) + \beta \delta(r)$$

Core Limit

$$r_D \rightarrow 0 \quad A_{\text{PMM}}(0) \text{ finite}$$

Exterior GR Limit

$$\delta \rightarrow 0 \quad r_D \gg r_{\text{core}}$$

$$A_{\text{PMM}}(r) \rightarrow 1 - 2G_N \frac{M}{(c^2 r_D)}$$

Closure-Stall Regularization

$\beta > 0$ Saturated δ raises the Θ_{out} threshold ($\alpha > 0$) until closure cannot fire; the FTU is held open rather than pushed outward. No z_{dir} sign flip.

Remnant Condition

$$F_{\text{tu}}^{\text{s}} \geq \hbar^3$$

$$N \geq 1$$

Mass Source Bridge

$$F_{\text{tu}}^{\text{Md}} = m$$

$$K_j = \kappa_G m_j (\kappa_G = 4\pi\hbar c/m_p^2 = 4\pi G_N, \text{ gravity bond scale})$$

$$\kappa_{\text{ij}}(r) = G_N m_j / r_{\text{ij}}^2$$

$$\theta_G = 45^\circ, \tan(\theta_G) = 1$$

$$T_{u\,ij}^{\text{G}} = G_N m_j \cdot \hat{n}_{ij} / r_{\text{ij}}^2$$

Universal Free-Fall Axiom

$T_u^{\text{G}} = G^{\text{N}} m_j / r^2$ is already an acceleration; test mass m_i never enters.

$$a_i^{\text{D}} = G_i^{\text{D}} = \Sigma_j T_{u\,ij}^{\text{G}} = \Sigma_j G^{\text{N}} m_j \hat{n}_{ij} / r_{\text{ij}}^2$$

Same D-location

$$a_i^{\text{D}} = a_k^{\text{D}} = a^{\text{G}}(r)$$

Meaning

The tug is a source-and-location acceleration; m_i does not appear. Free-fall depends on source m_j and separation r only. Body composition leaves the acceleration unchanged (weak equivalence, now structural rather than a cancellation of m_i).

PMM Statement

Black-hole remnant = an FTU held open by saturated- δ suppression of Θ_{out} , floored at one closure record ($N \geq 1$). Not a repulsive force; a stalled transcription.

2.4 14. MOND

The MOND block treats low-acceleration behavior as Φ -gradient behavior with a contact-write source.

Start with During

$$D = (c^1, c^2)_C$$

$$c^1 = \text{During traversal / time} - \text{gravity role}$$

$$c^2 = \text{relational realization / energy} - \text{mass role}$$

$$C = \text{light} - \text{speed causal gauge}$$

Depth-Modified During

$$D^\delta = (c^1, c^2)_{C^\delta}$$

D-Phase Metric

$$D^\delta \rightarrow g_{\mu\nu}^D(\Phi, \delta)$$

PMM Field Form

$$\nabla \cdot [\mu(|\nabla\psi|/a_0)\nabla\psi] = 4\pi G_N(\rho_b + \xi\Phi)$$

PMM Assignment

$$\Phi = \text{transported phase} - \text{memory record}$$

$$\psi = \text{gravitational readout potential}$$

$$\nabla\psi = \text{effective gravitational field}$$

$$\rho_b = \text{realized baryon density}$$

$$\xi\Phi = \text{carried record as effective source}$$

$$S_{\textcircled{a}} = \text{contact} - \text{write source}$$

a_0 = emergent acceleration scale

μ = interpolation function

Interpolation Rule

$$\mu(x) \approx 1 \text{ for } x \gg 1$$

$$\mu(x) \approx x \text{ for } x \ll 1$$

Acceleration Ratio

$$x = |\nabla\psi|/a_0$$

GR / Newtonian Limit

$$|\nabla\psi| \gg a_0$$

$$\mu(|\nabla\psi|/a_0) \approx 1$$

$$\nabla^2\psi \approx 4\pi G_N(\rho_b + \xi\Phi)$$

MOND-Like Limit

$$|\nabla\psi| \ll a_0$$

$$\mu(|\nabla\psi|/a_0) \approx |\nabla\psi|/a_0$$

$$\nabla \cdot [(|\nabla\psi|/a_0)\nabla\psi] \approx 4\pi G_N(\rho_b + \xi\Phi)$$

Rotation Curve Rule

$$g \approx \sqrt{G_N \text{Ma}_0}/r$$

Velocity Relation

$$\frac{v^2}{r} \approx \sqrt{G_N \text{Ma}_0}/r$$

Tully-Fisher Relation

$$v^4 \propto G_N \text{Ma}_0$$

PMM Source Map

$$S_{\textcircled{a}} = \Gamma_{\textcircled{a}}^{\Phi} \langle \Theta_{\text{out}} (\Delta T)_{\text{eff}}^2 \rangle$$

Emergent Scale

$$a_0^2 \sim \Gamma_{\textcircled{a}}^{\Phi} n_{\textcircled{a}} \langle (\Delta T)_{\text{eff}}^2 \rangle$$

Depth Field Link

$$\delta[\Phi,\rho] = \ell_\delta \cdot |\nabla^2 \Phi|/(|\nabla \Phi| + \epsilon_\Phi) \cdot \frac{\rho}{(\rho + \rho_c)}$$

One Line Bridge

$$D = (c^1, c^2)_{\mathcal{C}} \rightarrow D^\delta \rightarrow g_{\mu\nu}^{\mathcal{D}}(\Phi, \delta) \rightarrow \nabla \cdot [\mu(|\nabla \psi|/a_0) \nabla \psi] = 4\pi G_N(\rho_b + \xi \Phi)$$

2.5 15. Multi-FTU Tests

2.5.1 Tests for $a_0 \approx cH_0$

The purpose of the tests section is to show that PMM reaches $a_0 \sim cH_0/(2\pi)$ through a repeatable chain of horizon temperature, A-phase clock rate, and C-gauge acceleration, rather than by inserting a fitted galaxy-scale constant.

Consistency Checks for $a_0 \approx \text{cH}_0/2\pi$

$$T_H = \hbar H/(2\pi k_{\text{B}})$$

$$\gamma_H = k_{\text{B}} T_H/\hbar$$

$$\gamma_H = \frac{H}{2\pi}$$

$$a_H = c\gamma_H$$

$$a_H = \frac{cH}{2\pi}$$

Λ -only:

$$H_{\Lambda} = c/L_{\Lambda}$$

$$H_{\Lambda} = H_0 \sqrt{\Omega_{\Lambda}}$$

$$a_{\Lambda} = cH_{\Lambda}/2\pi$$

$$a_{\Lambda} \approx 8.6 \times 10^{-11} \frac{m}{s^2}$$

Full expansion:

$$a_{H0} = cH_0/2\pi$$

Present-epoch conversion:

$$a_{H0}/a_{\Lambda} = H_0/H_{\Lambda}$$

$$a_{H0}/a_{\Lambda} = 1/\sqrt{\Omega_{\Lambda}}, 0$$

Observed comparison:

$$a_{0, obs} \approx 1.16\text{--}1.20 \times 10^{-10} \frac{m}{s^2}$$

$$H_0 = 67.4 \frac{km}{s}/Mpc \Rightarrow a_{H0} \approx 1.04 \times 10^{-10} \frac{m}{s^2}$$

$$H_0 = 73.0 \frac{km}{s}/Mpc \Rightarrow a_{H0} \approx 1.13 \times 10^{-10} \frac{m}{s^2}$$

PMM read: Horizon temperature supplies the A-phase clock. γ_H supplies the horizon rate. The C-gauge reads γ_H as D-phase acceleration. a_{Λ} is the Λ -only horizon readout. a_{H0} is the full expansion horizon readout. $a_{0, obs}$ is the galaxy-fit comparison scale.

Main success: a_0 naturally lands near $cH_0/2\pi$ without an inserted fit factor.

2.5.2 Many-Bond Newtonian Recovery Test

Purpose: test whether multi-FTU contact-bond remainders recover Newtonian inverse-square gravity.

Universal Gravitational Crease

$$G = \text{universal FTU} - \text{to} - \text{FTU gravitational crease}$$

$$\theta_G = \text{angular measure of } G$$

$$\theta_G = 45^\circ$$

Pairwise Bond Tug

$$T_{u\,ij}^G = \tan(\theta_G) \kappa_{ij}(r) \hat{n}_{ij}$$

Area dilution

$$K_j = \text{total scalar bond supply of source } j$$

$$\oint \kappa_{ij}(r) dA = K_j$$

$$A(r) = 4\pi r^2$$

$$\kappa_{ij}(r) = K_j / 4\pi r^2$$

Many-bond readout

$$G_i^D = \Sigma_j T_{u\,ij}^G$$

$$G_i^D = \Sigma_j \tan(\theta_G) K_j \hat{n}_{ij} / 4\pi r_{ij}^2$$

Mass source limit

$$K_j = \kappa_G m_j, \kappa_G = 4\pi \hbar c / m_p^2 = 4\pi G_N$$

$$G_i^D = \Sigma_j \tan(\theta_G) \cdot G_N m_j \hat{n}_{ij} / r_{ij}^2 = \Sigma_j G_N m_j \hat{n}_{ij} / r_{ij}^2$$

Newtonian exterior limit

$$\text{For a spherical source with total mass } M : |G^D(r)| \propto \frac{M}{r^2}$$

$$\text{direction} \rightarrow \text{inward toward source center}$$

Shell check

Inside a symmetric shell:

$$\Sigma_j T_{u_{ij}}^G \rightarrow 0$$

while

$$\Sigma_j \kappa_{ij} > 0$$

Pass Condition

κ gives scalar source strength. $T_{u_{ij}}^G$ gives inward field direction. Many-bond composition gives $|G^D| \propto 1/r^2$ outside a spherical source. Symmetric shell geometry cancels vector G^D inside the shell.

3 Appendix A. Symbols and Constants

Central reference for the constants, parameters, and fields used above. Symbols are defined where first introduced; this section collects them.

3.1 Fundamental Constants

c = speed of light; D-phase causal gauge (C)

$\hbar$ = reduced Planck constant; B-D geometric crease and boundary relation; on-shell cube edge ($S^B = S^A = \hbar$)

m_p = Planck mass; B-axis scale carrier

k_B = Boltzmann constant; D-A geometric crease and boundary relation; A-axis scale carrier

ℓ_P = Planck length; $A_{\text{ref}} = 4\ell_P^2$

G = universal FTU-to-FTU gravitational crease and boundary relation

θ_G = angular measure of the G crease expressed through contact-generated per-vantage c^1/c^2 D-phase geometry

$\theta_G = 45^\circ$

$\tan(\theta_G) = 1$; relational gravity projection

$\sin(\theta_G) = 1/\sqrt{2}$; rail-normal depth projection

$\kappa_G = 4\pi\hbar c/m_p^2 = 4\pi G_N$; gravitational bond scale (magnitude carrier)

$\kappa_0 = c$; action-cube bond scale (distinct from κ_G)

$\tan(\theta^G) = G^N m_p / \ell_p c^2 = G^N m_p^2 / \hbar c = 1$ (Planck-mass identity, definitional)

$G^N = 6.67430 \times 10^{-11} \text{ m}^3 \cdot \text{kg}^{-1} \cdot \text{s}^{-2}$; Newton's constant

3.2 PMM Parameters and Scales

Local-vantage convention: unindexed m , d , L_0 , S^D , and z_{mag} belong to the active FTU vantage; i/j indices are reserved for relational expressions.

$L_0 = \hbar/(mc)$; local elementary on-shell y-rail span (c^2)

d = local traversal distance; c^1 /x-rail span; on-shell $d = L_0$

$z_{\text{mag}} = dL_0 \sin(\theta_G) = |F_{\text{tu}}^{\text{zd}}|$

$\theta_G = 45^\circ \Rightarrow z_{\text{mag}} = dL_0/\sqrt{2}$

on-shell $d = L_0 \Rightarrow z_{\text{mag}} = L_0^2/\sqrt{2}$

$A_{\text{ref}} = 4\ell_P^2$; c^3 -transcription reference area

$a_0^2 \sim \Gamma_{\text{@}}^\Phi n_{\text{@}} \langle (\Delta T)_{\text{eff}}^2 \rangle$; local emergent acceleration scale

ℓ_δ = depth length scale

ρ_c = critical density scale

ϵ_Φ = regulator in $(|\nabla\Phi| + \epsilon_\Phi)$

α = depth-gate coupling in Θ_{out}

α_δ = depth self-coupling in $\mathcal{L}_\delta$

Φ_{max} = phase-memory saturation bound

$\Delta S = S^D - S^B = mcd - \hbar$; off-shell action contrast ($= 0$ on-shell)

$(\Delta T)_{\text{eff}}^2 = (mcd/\hbar - 1)^2(1 - \Phi/\Phi_{\text{max}})$; dimensionless off-shell source factor

$\Gamma_{\text{@}}^\Phi$ = contact-write rate constant (carries the dimension balancing $\partial\Phi/\partial\tau_A$; see §7)

$n_{\text{@}}$ = contact-event number density

v_Φ = transport velocity/frame for Φ

$u_{\text{in}}, u_{\text{out}}$ = gate thresholds ($u_{\text{in}} < u_{\text{out}}$)

$\sigma_{\text{in}}, \sigma_{\text{out}}$ = gate widths

z_{dir} = inward/outward movement behavior

ψ = gravitational readout potential; $\nabla\psi$ = effective field

ρ_b = realized baryon density (source with $\xi\Phi$)

ξ = coupling of carried Φ to the gravitational source

u = local $B \rightarrow D \rightarrow A$ phase parameter

T_p = phase-memory persistence time; $T_p \rightarrow \infty$ = permanent

3.3 Core Fields and Operators

Σ = base orientation field/closure spread

Σ^D = optional two-rail spinor phase field used in $\Theta_{\text{con}}^{(\Sigma)}$

Φ = phase-memory field

δ = structural depth field; $\delta \notin B$, $\delta \in D$, $\delta \notin A$

@ = contact event opening FTU traversal/wave function collapse

$\mathcal{R}_i(c_j^1 \mid c_i^1) = c_{j \rightarrow i}^2$; defining per-vantage relational realization map

$\mathcal{R}_i$ adds no independent angular rotation or free scale; the receiving FTU supplies $L_{0i} = \hbar/(m \ c)$

c_i^1 = During traversal contribution of FTU_{*i*}

$c_{j \rightarrow i}^2$ = traversal contribution of FTU_{*j*} relationally realized from vantage i

c^1 -to- c^1 coupling establishes c^2 -order relational realization; $\mathcal{R}_i$ is the defining per-vantage operation

Θ_{in} = entry gate

Θ_{out} = closure gate

Φ_{n+1} = Memory update

$\Theta_{\text{win}} = \Theta_{\text{in}}[\Phi](1 - \Theta_{\text{out}}[\diamond])$; D-phase window gate (in, not yet out)

$\Theta_{\text{con}} = \{C_1[@, \Phi], C_2[\Phi, \delta, \Sigma], C_3[\Phi, \Sigma]\}$

$\mathcal{T}_A = c^3$ -transcription operator; $\mathcal{T}_A[\diamond] \rightarrow P(\Omega_a)$

$\diamond = (F_{\text{tu}}^y, F_{\text{tu}}^x \mid \delta, \Theta_{\text{win}}, \Theta_{\text{con}})$

$\mathbf{C}$ = D-phase rail gauge

$\mathbf{C}^\delta$ = depth-modified D-phase rail gauge

$\mathbf{N}$ = closure count per physical contact event; $\mathbf{N}(@_{ij}) = 1$

κ_{ij} = local contact-bond density from j available to i

K_j = total scalar bond supply of source j

$\hat{n}_{ij}$ = inward relational unit direction from i toward j

$T_{u \ ij}^G$ = pairwise directional bond expression (inward)

G_i^D = total During gravity readout at i

c_C = speed-valued carrier participation readout

$\chi = c_C/c = d\tau/dt$

$E_{\text{realization}}$ = D-phase energy realization readout

D^δ = During propagation mode through $\mathbf{C}^\delta$

$D^{c\delta}$ = lightlike During propagation mode

$\hat{t}^D$ = dimensionless unit tangent of D-flow

M_{op} = During-flow velocity operator

$M_{\text{op}}[D^\delta] = v_{\text{flow}}^D$

$M_{\text{op}}[D^{c\delta}] = c\hat{t}^D$

$M_{\text{op}}^c = \chi(v) E_{\text{realization}} z_{\text{dir}}$

$$[M_{\text{op}}^c] = E = J = \text{kg m}^2 \text{s}^{-2}$$

superscript c on M_{op}^c = carrier

$$T_{u,ij}^G = \tan(\theta_G) \kappa_{ij} \hat{n}_{ij}$$

$$G_i^D = \Sigma_j T_{u,ij}^G$$

The intrinsic FTU time-gradient readout is $F_{\text{tu}}^G = \nabla(F_{\text{tu}}^T)$.

G governs the universal gravitational crease between interacting FTUs; θ_G is its angular measure.

Observable gravity is expressed through the resulting inter-FTU bond relation.

3.4 Release Notes

3.4.1 2.7.0 - August 18, 2026

1. Derived c^2 as relational realization generated through coupling of two c^1 -order During contributions.
2. Defined $\mathcal{R}_i(c_j^1 \mid c_i^1) = c_{j \rightarrow i}^2$ as the canonical per-vantage realization map.
3. Restored the c-order geometric reading: c^0 point/scalar, c^1 edge, c^2 surface, c^3 volume, alongside the physical roles.
4. Specified $\mathcal{R}$ as adding no independent angular rotation or free scale; the receiving FTU supplies its local realization span $L_0 = \hbar/(\text{mc})$ under the local-vantage convention.
5. Recast per-vantage During as $D_i = (c_i^1, c_{j \rightarrow i}^2)\mathcal{C}$ and $D_j = (c_j^1, c_{i \rightarrow j}^2)\mathcal{C}$.
6. Kept local realization, depth, action, and SR readouts in compact unindexed form under the local-vantage convention, while reserving i/j indices for relational expressions and source mass m_j in the gravity bond.
7. Defined one physical contact $@_{ij}$ as one closure count, with A_i and A_j as two relational records of that one closure; per-vantage action readouts are never double-counted.
8. Separated common contact quantities from vantage-local depth readouts z_i and z_j .
9. Separated G as the universal FTU-to-FTU gravitational crease from θ_G as its angular measure; preserved $\theta_G = 45^\circ$, κ_G gravity scale, and Newtonian recovery.
10. Standardized the gravitational projection on $\tan(\theta_G)$ and reserved G_N for the dimensional Newtonian constant in standard equations.

3.4.2 2.6.9 - August 17, 2026

1. Defined fundamental constants as geometric creases expressing invariant boundary relationships.
2. Identified $\hbar$ as the B-D crease, k_B as the D-A crease, and G as the FTU-to-FTU gravitational crease, with θ_G as its 45° angular measure.

3. Made the dual D-phase rails explicitly relational: each contact couples self-D and partner-D perspectives.
4. Clarified traversal and realization as functional roles of the coupled rails from a given FTU vantage.
5. Preserved $\theta_G = 45^\circ$, $\tan(\theta_G) = 1$, $\sin(\theta_G) = 1/\sqrt{2}$, and the existing gravity, depth, SR, and Newtonian recovery mathematics.

3.4.3 2.6.8 - August 16, 2026

1. Defined θ^G as the angular measure used for the gravitational crease in the c^1/c^2 rail geometry.
2. Added $z_{\text{mag}} = dL_0 \sin(\theta^G)$.
3. Assigned $\tan(\theta^G)$ to gravity and $\sin(\theta^G)$ to depth projection.
4. Confirmed SR preserves the creased rail geometry and prior gravity recoveries.

3.4.4 2.6.7 - July 15, 2026

1. Split bond scales: $\kappa_0 = c$ (action) vs $\kappa_G = 4\pi G^N$ (gravity); $\kappa_0 = c$ cannot source gravity — wrong units by m^2 .
2. $\theta^G = 45^\circ$ orientation-only; $\tan(\theta^G) = 1$ via $G^N m_p^2 / \hbar c = 1$ (definitional).
3. Newtonian recovery closes exact: $|G^D| = \Sigma_j G^N m_j / r_{ij}^2$.
4. Retired the $(c^3/\hbar)$ bridge (units short by m^2).
5. Free-fall axiom corrected: $a_i^D = G_i^D$ (T_u^G is acceleration).

3.4.5 2.6.6 - July 15, 2026

1. Modified spinor extension behavior to capture dual rail; $2\pi/4\pi$ symmetry labeled postulated
2. Reverted header gravity claim to c^1 parity form; $B \circ D \circ A \rightarrow \epsilon$ chain remains a derivation path, not a result
3. Unified $q_{\text{retained}} \rightarrow T_{u\,ij}^G$; added bare- T_u^G shorthand convention
4. Single direction account restored: $\nabla\kappa$ supplies inward direction
5. Bridge conditional added: $v_{\text{flow}}^D = v_{xy}$ massive only; photon v_{xy} undefined
6. Proposed-reading tags added to double-slit and EM light blocks
7. Regulator ϵ renamed ϵ_Φ to close collision with rail-parity $R_{\text{as},ij}$

3.4.6 2.6.5 - July 14, 2026

1. Separated During propagation mode D^δ from velocity operator M_{op} .
2. Defined $M_{\text{op}}[D^\delta]$ as a velocity readout with units $\text{L} \cdot T^{-1}$.
3. Defined M_{op}^c as carrier participation energy with units E.
4. Standardized c_C as the speed-valued carrier participation readout.
5. Moved z_{dir} into the gate-derived orientation readout.
6. Tagged the photon-marker picture as a proposed PMM reading.

3.4.7 2.6.4 - July 11, 2026

1. G^D demoted from primitive to derived theorem of $F_{\text{tu}}^G = \nabla(F_{\text{tu}}^T)$ via bond quantization ($K \cdot L_0 = \hbar \Rightarrow K_j = m_j c, \kappa_0 = c$ forced)
2. $1/r^2$ and source-ward $\hat{n}$ derived, not posited
3. ϵ flagged as remaining primitive input, derivation pending
4. Restores single-primitive gravity telos: gravity writes the record, the record dilutes over the closure sphere as $1/r^2$, that dilution is G^D

3.4.8 2.6.3 - July 10, 2026

1. More clarity on rail bond asymmetry

3.4.9 2.6.2 - July 9, 2026

1. Multi-FTU gravity modification
2. Add many-bonds Newtonian gravity test results

3.4.10 2.6.1 - July 8, 2026

1. Added contact primitive
2. Added two vantage point contact

3.4.11 2.6.0 - Jul 7 2026

1. Added force readout
2. Depth set rail-normal: D-metric coordinates (ct,x,y); $z = c^1 \times_r c^2$ generated, not a coordinate
3. Momentum split — translational $p_{\text{xy}} \in \text{span}\{x, y\}$; rail-normal carrier participation readout M_{op}^c

4. Added motion readout section

3.4.12 2.5.9 - Jul 6 2026

1. Added momentum readout
2. Added D-Velocity Rule in Kinematics

3.4.13 2.5.8 - Jul 4 2026 (!USA 250!)

1. Modified to three acceleration readouts
2. Added a separate tests section

3.4.14 2.5.7

1. Closed off-shell variance $\langle \epsilon^2 \rangle = mc^2/k_B T$; a_0 temperature-independent, $a_0^2 \sim n_{@} mc^2/\hbar$, cosmological scale of $n_{@}$ flagged open.
2. FTU entropy readout section added

3.4.15 2.5.6

1. Closed $(\Delta T)_{\text{eff}}^2$ as off-shell action contrast $(mcd/\hbar - 1)^2$; dimensionless, vanishes on-shell, reconciled with B-D stress contrast.

3.4.16 2.5.5

1. Updated memory update form Φ_{n+1}

3.4.17 2.5.4

1. Added a clarified Φ definition field for completed FTUs.

3.4.18 2.5.3

1. Added Spatial Coordinate Readout Summary section

3.4.19 2.5.2

1. Added Lagrangian Section
2. Electromagnetic Extension

3.4.20 2.5.1

1. Clearer metric definitions
2. Explicit gravity subsection
3. Depth regulation test

3.4.21 2.5.0

1. Nomenclature and symbol cleanup
2. Added Action Readouts section

3.4.22 2.4.9

1. Upgraded Kinematics
2. Updated the SR section to reflect the new Kinematics notation
3. Updated Appendix with new Kinematics symbol